Windows Server 2025

Complete Installation, Configuration, Administration, Security & Troubleshooting Guide

A step-by-step successor to the district manual *Windows Server 2022 – Installation and Configuration*. Written so a first-time administrator can follow every procedure, with the depth an experienced administrator needs.

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Documentation-verified draft — not lab-tested

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VERSION	DATE	CHANGES
2022 edition	—	Original manual: Windows Server 2022 installation and configuration (115 pages).
1.0 Draft	Oct 15, 2026	Full rewrite for Windows Server 2025. Every original chapter audited, corrected and modernized; insecure advice replaced (IPv6, firewall, .local, automatic approvals, FRS); new chapters on Windows LAPS, hardening, Server Core and recovery. Remaining expansion chapters tracked in Appendix I.

Who this guide is for

Beginner and intermediate Windows Server administrators, K–12 technology staff, IT students and instructors — plus enterprise administrators who want a checked reference. You should be comfortable using Windows, typing commands, and understand what an IP address is. Everything else is explained when it first appears. Sections marked **Advanced** can be skipped on a first read.

How to read a procedure

1. **Read the whole procedure first.** Then do it step by step.
2. Most tasks have two methods: **GUI** (click-through, best for learning) and **PowerShell** (best for repeating the task on many servers). Use one or the other — not both.
3. Text in `this font` is something you type exactly, except values in angle brackets like `<ServerName>`, which you replace with your own value.
4. Every chapter ends with **Verify** steps. Do not skip them — they tell you whether the change actually worked.
5. Open PowerShell as administrator: right-click **Start** → **Terminal (Admin)** or **Windows PowerShell (Admin)**.

Warning labels

LABEL	MEANING
Security risk	The setting weakens or strengthens protection. Read before changing.
Destructive	Deletes data or objects and cannot easily be undone. Have a tested backup first.
Production impact	May interrupt users or services. Do it in a maintenance window.
Restart required	The change only takes effect after a reboot.
Admin rights	Requires local Administrator, Domain Admins, Enterprise Admins or Schema Admins, as stated.
Lab only	Safe for practice networks only; never run against production.
Organization-specific	An example from the original K–12 (Arkansas DIS) environment — not a universal requirement.

Validation status — read this

> **HONEST STATUS** Every procedure in this edition has been checked against Microsoft documentation (status: *Documentation verified*). **None has yet been executed in a Windows Server 2025 lab**, so nothing is labelled *Lab tested*. Items Microsoft documents inconsistently, or that vary by build, are flagged *Requires verification* in the text. Run every procedure in a test lab before production use. Screenshots from the 2022 edition were not reused; boxes marked "Screenshot required" show where an authentic Windows Server 2025 capture must be added.

The lab environment used in every example

All examples use reserved documentation addresses (RFC 5737) and the reserved `example.com` namespace, so nothing collides with a real network. Replace them with your own values.

ROLE	NAME	IPV4
Domain (DNS / NetBIOS)	`ad.example.com` / `AD`	192.0.2.0/24
First domain controller (DNS, DHCP)	`DC01.ad.example.com`	192.0.2.10
Second domain controller	`DC02.ad.example.com`	192.0.2.11
Member server (files, WSUS)	`SRV01.ad.example.com`	192.0.2.20
Client	`CLIENT01.ad.example.com`	DHCP (192.0.2.100–200)
Default gateway	—	192.0.2.1

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A–I	Appendices A–I: commands, ports, checklists, legacy technology, decision trees, glossary, references, traceability, status

1 · What's new in Windows Server 2025

Overview. Windows Server 2025 is the Long-Term Servicing Channel (LTSC) release that follows Windows Server 2022. It became generally available on November 1, 2024. If you know Windows Server 2022, most tools look the same — Server Manager, Active Directory Users and Computers, DNS Manager — but several security defaults changed. Those defaults are the most common cause of "it worked on 2022" problems.

You will learn: which changes affect everyday administration, which features depend on edition, hardware or Azure, and which older features are deprecated.

Changes that affect day-to-day administration

AREA	WHAT CHANGED	DEPENDS ON
AD DS functional level	New *Windows Server 2025* domain and forest functional level (the first since 2016). Raising a functional level is no longer one-way: you can lower it again to a supported level.	All DCs must run 2025 to use the 2025 level
Adding 2025 DCs	The existing domain and forest must already be at the *Windows Server 2016* functional level, and SYSVOL must replicate with DFS Replication (not FRS).	—
AD database	Optional 32k database page size feature (larger multi-valued attributes). Enabling it is irreversible.	2025 forest functional level
Service accounts	New *delegated Managed Service Accounts* (dMSA). Restrict who can create them — see Chapter 15.	2025 DCs
SMB signing	The SMB **client** now requires signing for all outbound connections by default. Inbound (server-side) signing is still required by default only on domain controllers. Older NAS devices that cannot sign will fail to connect.	—
SMB other	Insecure guest access blocked; SMB over QUIC available in Standard and Datacenter (previously Azure Edition only); SMB client can block NTLM; authentication rate limiter.	QUIC needs a TLS certificate

AREA	WHAT CHANGED	DEPENDS ON
Security baseline	Microsoft ships the 2025 baseline through *OSConfig*, a PowerShell module with drift control (Chapter 15).	—
Hotpatching	Install many security updates without rebooting.	Azure Arc + paid subscription (Standard/Datacenter); included in Azure Edition
Windows LAPS	Built in. Adds automatic account management and passphrase options.	—
Hyper-V	GPU partitioning with live migration, much larger VM maximums, workgroup clusters.	Edition, hardware
Upgrades	In-place upgrade from Windows Server 2012 R2, 2016, 2019 and 2022. For domain controllers, Microsoft's preferred method is to add new 2025 DCs and retire old ones.	—

Deprecated or removed — what it means for this guide

FEATURE	STATUS	REPLACEMENT USED IN THIS GUIDE
WINS	Deprecated; 2025 is the last LTSC release that includes it (supported through Nov 2034).	DNS (Appendix D keeps the old procedure)
WSUS	Deprecated — still included and supported, no new features. Driver sync deprecated.	Staged WSUS (Ch. 9) or Azure Update Manager / Configuration Manager
Windows Internal Database	Deprecated.	SQL Server for large WSUS deployments
VBScript	Deprecated; becoming a Feature on Demand before removal.	Group Policy Preferences, PowerShell (Ch. 10)
WMIC	Disabled by default on new installations.	PowerShell CIM cmdlets
NTLM (all versions)	Deprecated; NTLMv1 removed.	Kerberos
TLS 1.0 / 1.1	Disabled by default.	TLS 1.2 / 1.3
FRS for SYSVOL	Removed since Windows Server 2019.	DFS Replication

References: Microsoft Learn — *What's new in Windows Server 2025*; *Features removed or no longer developed in Windows Server*; *SMB security hardening*; KB5071836 *WINS removal*.

2 · Requirements, editions, licensing & lifecycle

Overview. Before buying hardware or downloading media, confirm the server meets the minimums, choose an edition, and work out how many core licenses and CALs you need.

2.1 Minimum hardware

These are Microsoft's *minimums* to install. Real servers need more — size for the roles you will run.

COMPONENT	MINIMUM	PRACTICAL RECOMMENDATION
Processor	1.4 GHz 64-bit; must support NX, DEP, CMPXCHG16b, LAHF/SAHF, PrefetchW, **SSE4.2 and POPCNT** (new requirements that rule out some very old CPUs). Hyper-V also needs SLAT (EPT/NPT).	A processor on your server vendor's Windows Server 2025 certified list
Memory	512 MB (Server Core) · 2 GB (Desktop Experience). ECC for physical hosts.	8 GB+ for a domain controller; more for file, WSUS or Hyper-V
Storage	32 GB — an absolute minimum that leaves no room for updates	100–150 GB system volume (C:). Put data, WSUS content and home folders on separate volumes.
Network	Ethernet adapter, 1 Gbps	Two adapters or a team for hosts; static IPv4 for servers
Firmware	UEFI 2.3.1c with Secure Boot recommended	Leave **Secure Boot on**
TPM	Not required to install	TPM 2.0 — needed for BitLocker with TPM, Credential Guard/VBS and Secured-core
Power	—	UPS with surge protection; configure graceful shutdown

> **NOTE** The 2022 manual listed "Super VGA (800×600)" and a DVD drive. Neither matters today — install from USB or a virtual ISO, and manage servers remotely.

2.2 Editions

EDITION	BEST FOR	VIRTUALIZATION RIGHTS (ALL PHYSICAL CORES LICENSED)
Standard	Physical servers or lightly virtualized hosts	2 OSEs (VMs) or Hyper-V containers; unlimited Windows Server containers
Datacenter	Heavily virtualized hosts, Storage Spaces Direct, Storage Replica without limits, SDN	Unlimited OSEs and Hyper-V containers
Datacenter: Azure Edition	Runs only on Azure and Azure Local (formerly Azure Stack HCI); includes hotpatching	—
Essentials	Very small businesses through OEM partners (limited users/devices)	—

Both Standard and Datacenter can be installed as **Server Core** (no desktop; smaller, fewer updates, smaller attack surface) or **Desktop Experience** (full GUI). You cannot switch between them after installation — reinstall instead. Beginners may start with Desktop Experience; production roles such as domain controllers are good Server Core candidates once you are comfortable with remote tools (Section 4.9).

2.3 Licensing

- **Core licensing:** license every physical core, with a minimum of **8 cores per processor** and **16 cores per server**.
- **Standard stacking:** each additional full set of core licenses adds two more OSEs. Example: a 2-processor server with 8 cores each needs 16 core licenses for 2 VMs; licensing the same 16 cores again gives 4 VMs.
- **CALs:** every user or device that accesses the server needs a Windows Server CAL (user or device). Remote Desktop Services needs separate RDS CALs.
- **Pay-as-you-go:** Windows Server 2025 can also be licensed per core through Azure Arc, billed monthly to an Azure subscription — check current Microsoft Product Terms.
- **Virtual machine licensing** (with Software Assurance or subscription): license per VM, minimum 8 cores per VM.

> **ORGANIZATION-SPECIFIC** Licensing terms change. Confirm with Microsoft Product Terms, your reseller or your state/education agreement before purchasing. Volume licensing downloads and keys moved from the retired Volume Licensing Service Center (VLSC) to the *Microsoft 365 admin center*.

2.4 Support lifecycle

Mainstream support brings fixes and improvements; extended support brings security updates only. Plan upgrades well before the extended date.

VERSION	MAINSTREAM ENDS	EXTENDED (SECURITY) ENDS
Windows Server 2025	Nov 13, 2029	Nov 14, 2034
Windows Server 2022	Oct 13, 2026 (ended)	Oct 14, 2031
Windows Server 2019	Jan 9, 2024 (ended)	Jan 9, 2029
Windows Server 2016	Jan 11, 2022 (ended)	Jan 12, 2027
Windows Server 2012 / 2012 R2	Ended	Ended Oct 10, 2023; paid ESU ended Oct 13, 2026

> **CORRECTION** The 2022 manual labelled extended-support dates as "mainstream" (e.g. 2016 "mainstream January 12, 2027"). The table above separates them. Source: Microsoft Lifecycle product pages, reviewed October 2026.

3 · Installation media & installing Windows Server 2025

Overview. You will create a bootable USB drive that works with UEFI and Secure Boot left on, then install Windows Server 2025 on a clean disk.

Prerequisites: Windows Server 2025 ISO (Microsoft 365 admin center for volume licensing, or the Microsoft Evaluation Center for a 180-day evaluation); a USB drive of 16–32 GB that you can erase; storage controller (RAID) drivers if your server vendor requires them; RAID arrays/logical drives already created in the controller's setup utility; a planned static IP address and server name.

3.1 Why the USB drive needs special handling

UEFI firmware boots from a **FAT32** partition, and FAT32 cannot hold files larger than 4 GB. The Windows Server image file `sources\install.wim` is usually larger than 4 GB. The fix is to **split** that file — not to disable Secure Boot or switch to legacy BIOS mode, as older guides suggested.

3.2 Create the USB drive — GUI method

Use your server vendor's tool or a well-known utility such as Rufus. Choose: partition scheme **GPT**, target system **UEFI (non-CSM)**, file system as recommended by the tool. Download such tools only from the publisher's official site and verify the ISO's SHA-256 hash against Microsoft's published value when available.

3.3 Create the USB drive — PowerShell method

> **DESTRUCTIVE** These commands erase the selected disk. Check the disk number and size twice. Unplug other USB drives first.

1. Insert the USB drive and open PowerShell as administrator.
2. Identify the USB disk:

   ```
   Get-Disk | Where-Object BusType -eq 'USB' | Format-Table Number, FriendlyName, Size
   ```

3. Erase it and create one 32 GB-or-smaller FAT32 partition (Windows formats FAT32 only up to 32 GB). Replace `<N>` with the disk number:

```
$disk = Get-Disk -Number <N>
$disk | Clear-Disk -RemoveData -RemoveOEM -Confirm:$true
$disk | Initialize-Disk -PartitionStyle GPT
$size = [Math]::Min($disk.Size - 64MB, 32GB)
$part = New-Partition -DiskNumber $disk.Number -Size $size -AssignDriveLetter
Format-Volume -Partition $part -FileSystem FAT32 -NewFileSystemLabel 'WS2025'
$usb = ($part | Get-Partition).DriveLetter + ':'
```

4. Mount the ISO and copy everything except `install.wim`:

```
$iso = Mount-DiskImage -ImagePath 'C:\ISO\WS2025.iso' -PassThru
$src = ($iso | Get-Volume).DriveLetter + ':'
robocopy $src\ $usb\ /E /XF install.wim /NFL /NDL
```

5. Split `install.wim` into pieces under 4 GB (Setup reads `.swm` files automatically):

```
dism /Split-Image /ImageFile:$src\sources\install.wim /SWMFile:$usb\sources\install.swm
/FileSize:3800
Dismount-DiskImage -ImagePath 'C:\ISO\WS2025.iso'
```

Verify: `Get-ChildItem $usb\sources\install*.swm` lists two or more files, and `$usb\efi\boot\bootx64.efi` exists.

3.4 Install Windows Server 2025

Windows Server 2025 uses the newer Windows Setup interface. Screen titles below match current media; confirm against your build.

1. Insert the USB drive, power on, and open the one-time boot menu (often **F11** or **F12**). Choose the **UEFI** entry for the USB drive.
2. **Select language settings** and **keyboard** → **Next**.
3. **Select setup option:** choose *Install Windows Server*, tick the confirmation that everything on the target will be deleted, → **Next**.
4. **Product key:** enter your key, or choose *I don't have a product key* to activate later.
5. **Select image:** pick the edition. Entries ending in *(Desktop Experience)* include the GUI; the others install Server Core.
6. Accept the license terms.
7. **Select location to install:** if the disk is not listed, choose *Load driver* and supply the RAID driver. Select the *unallocated space* and click **Next** — Setup creates the EFI, MSR, Windows and recovery partitions correctly on a GPT disk.

To create a specific size instead, choose *Create partition* and enter the size in MB (1 GB = 1024 MB; 150 GB = 153600 MB).

8. **Ready to install** → **Install**. The server restarts several times. Remove the USB drive after the first restart if it boots back into Setup.
9. Set the local **Administrator** password, then sign in with **Ctrl**+**Alt**+**Del**.

> Screenshot required: Windows Server 2025 Setup — "Select image" screen showing Standard / Datacenter with and without Desktop Experience.

> **SECURITY RISK** The default policy requires at least 7 characters with three of four character types, but that is a floor, not a goal. Use a long, unique passphrase (15+ characters) for the built-in Administrator, store it in your password vault, and never reuse examples such as *Password1*. After joining a domain, Windows LAPS (Chapter 14) manages this password automatically.

Correction from the 2022 edition: the size formula "MB × 10240" was wrong; and partition style is chosen automatically — UEFI systems always use GPT.

Troubleshooting: *No disks shown* → load the storage driver or check RAID configuration. *"Windows cannot be installed to this disk… MBR"* → you booted in legacy mode; reboot and select the UEFI USB entry. *Setup fails reading install.swm* → recreate the USB drive; the split did not complete.

4 · Initial server configuration

Overview. A freshly installed server needs a name, a static IP address, correct time zone, activation, remote management and updates before it takes on any role. Do these steps in the order shown — renaming after promoting a domain controller is far harder.

Prerequisites: local Administrator sign-in; a naming convention; the IP address, subnet, gateway and DNS server values for this server.

4.1 Open Local Server

Server Manager starts automatically. Click **Local Server** in the left pane. Each blue link (Computer name, Ethernet, Time zone, Product ID…) opens the related setting. Alternatively run `sconfig` — a menu-driven tool available in both Server Core and Desktop Experience.

4.2 Rename the server

Use a convention that identifies site and role, for example `HS-DC01` (high school, domain controller 1) or `ADM-FS01`. Maximum 15 characters; letters, digits and hyphens only.

GUI: Local Server → click the *Computer name* → **Change…** → type the name → **OK** → restart.

```
Rename-Computer -NewName 'DC01' -Restart
```

4.3 Configure a static IP address

GUI: Local Server → click the *Ethernet* link → right-click the adapter → **Properties** → *Internet Protocol Version 4 (TCP/IPv4)* → **Properties** → enter IP, subnet mask, gateway and DNS servers.

```
Get-NetAdapter                       # note the InterfaceAlias, e.g. "Ethernet"
New-NetIPAddress -InterfaceAlias 'Ethernet' -IPAddress 192.0.2.10 -PrefixLength 24 -
DefaultGateway 192.0.2.1
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses 192.0.2.11,192.0.2.10
```

Which DNS servers? Domain members point only at your domain controllers — never at public DNS. For a domain controller's own DNS settings see Section 5.3.

4.4 IPv6 — leave it enabled

> **CORRECTION** The 2022 manual told readers to untick IPv6 and set the registry value `DisabledComponents = ffffffff`. Microsoft does not recommend disabling IPv6: Windows is designed and tested with IPv6 on, some components (including parts of Active Directory, failover clustering and DirectAccess) expect it, and the value `0xFFFFFFFF` also causes a startup delay. Leaving IPv6 enabled does *not* expose a server to the internet — your firewall and network design control that.

If a specific application misbehaves and you need IPv4 to be used first, **prefer IPv4 over IPv6** instead of disabling it:

```
# Prefer IPv4 over IPv6 (Microsoft-documented value 0x20). Restart required.
New-ItemProperty -Path 'HKLM:\SYSTEM\CurrentControlSet\Services\Tcpip6\Parameters' `
  -Name DisabledComponents -PropertyType DWord -Value 0x20 -Force
Restart-Computer
```

Verify: `ping DC01` should now reply from the IPv4 address. `Get-NetPrefixPolicy` shows the IPv4-mapped prefix (`::ffff:0:0/96`) with the highest precedence. **Rollback:** delete the `DisabledComponents` value and restart.

4.5 Time zone

```
Set-TimeZone -Id 'Central Standard Time'      # list options: Get-TimeZone -ListAvailable
```

Time accuracy matters more than you might think: Kerberos authentication fails when clocks differ by more than 5 minutes. Chapter 8 configures the domain time hierarchy.

4.6 Activate Windows

GUI: Local Server → *Product ID* → enter the key → **Next** → **Activate**. Requires internet access.

Command line (Command Prompt or PowerShell, as administrator):

```
slmgr.vbs /ipk XXXXX-XXXXX-XXXXX-XXXXX-XXXXX   # install key
slmgr.vbs /ato                                  # activate online
slmgr.vbs /dlv                                  # show detailed license status
```

Other activation methods: **KMS** or **Active Directory-Based Activation** for many servers with volume keys; **AVMA** for VMs on an activated Datacenter Hyper-V host; **Azure Arc** for pay-as-you-go. Evaluation installs convert with `DISM /Online /Set-Edition:ServerStandard /ProductKey:<key> /AcceptEula`.

Requires verification: `slmgr.vbs` runs on VBScript, which is deprecated. It works on current Windows Server 2025 builds; check Microsoft's VBScript timeline before relying on it in automation.

4.7 Remote management: Remote Desktop, WinRM and the firewall

1. **Remote Desktop:** Local Server → *Remote Desktop* → *Allow remote connections to this computer*, and keep *Allow connections only from computers running Remote Desktop with Network Level Authentication* ticked.

   ```
   Set-ItemProperty 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name
   fDenyTSConnections -Value 0
   Set-ItemProperty 'HKLM:\System\CurrentControlSet\Control\Terminal Server\WinStations\RDP-
   Tcp' -Name UserAuthentication -Value 1
   Enable-NetFirewallRule -DisplayGroup 'Remote Desktop'
   ```

2. **PowerShell remoting / Server Manager remote management** is on by default for domain and private networks. Check with `Test-WSMan DC01`.

> **SECURITY RISK** Never publish RDP (TCP/UDP 3389) directly to the internet. Use a VPN or Remote Desktop Gateway, and limit the RDP firewall rule to your management network (Section 4.8).

4.8 Windows Defender Firewall — keep it on

> **CORRECTION** The 2022 manual showed how to turn the firewall off for the Domain and Private profiles. Do not. A perimeter firewall does not stop an infected computer *inside* your network from attacking servers. Windows roles create the firewall rules they need automatically when installed; your job is to scope and log them.

Concepts: Windows uses three *profiles* — **Domain** (the network can reach a domain controller), **Private** and **Public**. Each has its own rules. *Inbound* rules control connections to the server; *outbound* rules control connections from it (allowed by default).

GUI: Server Manager → **Tools** → *Windows Defender Firewall with Advanced Security*. Use *Inbound Rules* to view, enable, and open a rule's *Scope* tab to restrict remote IP addresses.

```
# Confirm all profiles are on and log dropped packets
Get-NetFirewallProfile | Format-Table Name, Enabled, DefaultInboundAction
Set-NetFirewallProfile -Profile Domain,Private,Public -Enabled True -LogBlocked True -
LogMaxSizeKilobytes 16384

# Allow RDP only from the management subnet
Get-NetFirewallRule -DisplayGroup 'Remote Desktop' | Set-NetFirewallRule -RemoteAddress
192.0.2.0/27

# Example custom rule for an application port
New-NetFirewallRule -DisplayName 'App - TCP 8443 from app subnet' -Direction Inbound `
  -Protocol TCP -LocalPort 8443 -RemoteAddress 192.0.2.64/26 -Action Allow -Profile Domain
```

Verify: from another computer, `Test-NetConnection DC01 -Port 3389` succeeds from the management subnet and fails from elsewhere. The log is at `%SystemRoot%\System32\LogFiles\Firewall\pfirewall.log`.

Manage at scale with Group Policy: Computer Configuration → Policies → Windows Settings → Security Settings → Windows Defender Firewall with Advanced Security.

4.9 Server Core and SConfig (overview)

Server Core has no Start menu or Server Manager window. You sign in to a command prompt and use `sconfig` — a numbered menu for the same tasks covered in this chapter (domain join, computer name, network settings, updates, Remote Desktop, activation, telemetry, restart). Day-to-day you manage Server Core remotely with Server Manager, Remote Server Administration Tools (RSAT) or Windows Admin Center from an administrative workstation. Every PowerShell command in this guide works on Server Core.

```
Add-Computer -DomainName ad.example.com -Credential AD\admin.jsmith -Restart   # join a domain
Install-WindowsFeature DNS -IncludeManagementTools                             # install a role
Get-WindowsFeature | Where-Object Installed                                    # list installed
roles
```

4.10 Install updates

GUI: Settings → *Windows Update* → **Check for updates**. Repeat until no updates remain, restarting as asked. On Server Core use `sconfig` option 6. After joining the domain, updates are controlled by policy (Chapter 9).

4.11 Web browsing on servers

The 2022 manual turned off *IE Enhanced Security Configuration*. Windows Server 2025 uses Microsoft Edge, and the right practice is not to browse the web from servers at all: download tools on an administrative workstation, verify them, and copy them over. *Requires verification:* if your build still shows an "IE Enhanced Security Configuration" item in Local Server, leave it **On**.

4.12 Shutdown Event Tracker

The tracker asks why a server is being shut down and records the answer in the System log (event 1074). That record is useful for audits, so leave it on in production. If your organization decides to disable it, use Group Policy rather than hand-editing the registry: Computer Configuration → Policies → Administrative Templates → System → *Display Shutdown Event Tracker* → **Disabled**.

Verify this chapter

```
Get-ComputerInfo -Property CsName, WindowsProductName, OsVersion, TimeZone
Get-NetIPConfiguration
Get-NetFirewallProfile | Select-Object Name, Enabled
cscript //nologo C:\Windows\System32\slmgr.vbs /xpr     # activation status
```

Practical lab 4 — Prepare DC01 (Beginner, 30 min)

Objective: take a fresh VM to a named, addressed, activated, firewalled server. **Starting state:** Windows Server 2025 Desktop Experience just installed on a VM connected to an isolated virtual switch.

1. Rename the server to `DC01` and restart.
2. Set IPv4 192.0.2.10/24, gateway 192.0.2.1, DNS 127.0.0.1 (temporary until AD DS is installed).
3. Set the time zone and enable Remote Desktop with NLA.
4. Scope the Remote Desktop firewall rules to 192.0.2.0/24 and enable dropped-packet logging.
5. Run the verification commands above.

Expected result: `Get-NetIPConfiguration` shows 192.0.2.10; all firewall profiles report *True*. **Cleanup:** none — DC01 is used in Lab 5.

Knowledge check. 1) Why prefer IPv4 instead of disabling IPv6? *Disabling is unsupported in practice and breaks components; preferring IPv4 changes only address selection.* 2) Which firewall profile applies to a server that can reach a domain controller? *Domain.* 3) Why should RDP never face the internet? *It is a prime target for password-guessing and exploits; use VPN or RD Gateway.*

5 · Active Directory Domain Services

Overview. Active Directory Domain Services (AD DS) stores users, computers and groups, and authenticates sign-ins. A server running AD DS is a **domain controller (DC)**. The first DC creates a **forest** (the security boundary) containing your first **domain**. Every domain should have **at least two DCs** so sign-in keeps working when one fails.

You will learn to choose a domain name, install AD DS, create a new forest, add a second DC and check health.

Prerequisites: a server prepared as in Chapter 4 with a static IP; local Administrator rights (new forest) or Domain Admins (additional DC); a decided domain name; a DSRM password stored in your vault.

5.1 Choose the domain name

> **CORRECTION** The 2022 manual recommended names like `school.local`. Microsoft's current guidance is to use a **subdomain of a DNS name your organization owns**, for example `ad.district.k12.ar.us` or `corp.example.com`. Reasons: `.local` is reserved for multicast DNS (used by printers and Apple devices), public certificate authorities will not issue certificates for it, and it complicates hybrid identity with Microsoft Entra ID. Existing `.local` domains remain supported — do not rename a working domain just for this.

- Use a subdomain (`ad.`, `corp.`) — not your public website name itself — so internal and public DNS do not clash.
- The NetBIOS name (pre-Windows 2000 name) is up to 15 characters, e.g. `AD` or `DISTRICT`. Users may see it in `DISTRICT\username`.
- Avoid single-label names (`district`) and names you cannot prove ownership of.

5.2 Functional levels

FUNCTIONAL LEVEL	DCS ALLOWED	NOTES
Windows Server 2016	2016, 2019, 2022, 2025	Minimum for a new forest; required before adding a 2025 DC to an existing domain. 2019 and 2022 never had their own level.
Windows Server 2025	2025 only	Default for an all-2025 forest; enables 2025-only features such as the 32k database page size option.

Raising a level is now reversible: a domain containing 2025 DCs can be lowered back to 2016 with `Set-ADDomainMode` / `Set-ADForestMode`, *unless* a level-specific feature has already been enabled. Choose the 2016 level if 2016–2022 DCs still exist or will be added.

5.3 DNS client settings on domain controllers

- **First DC in a new forest:** preferred DNS = its own IP (192.0.2.10); alternate = `127.0.0.1`.
- **Additional DC before promotion:** preferred DNS = an existing DC (192.0.2.10). After promotion: preferred = a partner DC, alternate = its own IP, and loopback last.
- Never list public DNS servers (ISP, 8.8.8.8, filtering services) on a DC or domain member — use DNS forwarders instead (Chapter 6).

5.4 Install the AD DS role and create a new forest

GUI

1. Server Manager → **Manage** → *Add Roles and Features* → **Next** → *Role-based or feature-based installation* → select this server.
2. Tick **Active Directory Domain Services** → **Add Features** → **Next** through the remaining pages → **Install**. The DNS Server role is added during promotion; you do not need to install it first.
3. When installation finishes, click the **notifications flag** → *Promote this server to a domain controller*.
4. **Deployment Configuration:** *Add a new forest*; Root domain name `ad.example.com`.
5. **Domain Controller Options:** forest and domain functional level *Windows Server 2025* (or 2016 — see 5.2); keep *DNS server* and *Global Catalog* ticked; enter the **DSRM password** — unique to DCs and stored in your vault.
6. **DNS Options:** a warning that a delegation cannot be created is expected for a new internal namespace → **Next**.
7. **Additional Options:** confirm the NetBIOS name. **Paths:** accept defaults (`C:\Windows\NTDS`, `C:\Windows\SYSVOL`) unless you have a separate data volume.
8. **Review Options** → **View script** saves the equivalent PowerShell — a good habit. **Prerequisites Check** → read warnings → **Install**. The server restarts automatically.

> Screenshot required: Active Directory Domain Services Configuration Wizard — Domain Controller Options page on Windows Server 2025.

PowerShell

```
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
Install-ADDSForest -DomainName 'ad.example.com' -DomainNetbiosName 'AD' -InstallDns `
  -SafeModeAdministratorPassword (Read-Host -AsSecureString 'DSRM password')
# Functional level defaults to Windows Server 2025 on an all-2025 forest.
# Add -ForestMode WinThreshold -DomainMode WinThreshold for the 2016 level.
```

> **RESTART REQUIRED** After restart, sign in as `AD\Administrator` (the local Administrator becomes the domain Administrator). Create a separate named admin account immediately (Chapter 10) and stop using the built-in account day to day.

5.5 Add a second domain controller

1. Prepare DC02 (Chapter 4) with DNS pointing to DC01, and join it to the domain.
2. Install the AD DS role as above, then promote: *Add a domain controller to an existing domain*; supply Domain Admin credentials; keep DNS and Global Catalog ticked; choose the site; *Replicate from:* DC01.

```
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
Install-ADDSDomainController -DomainName 'ad.example.com' -InstallDns `
  -Credential (Get-Credential AD\admin.jsmith) `
  -SafeModeAdministratorPassword (Read-Host -AsSecureString 'DSRM password')
```

After it restarts, set DC02's DNS to 192.0.2.10, 192.0.2.11, 127.0.0.1 and add DC02 to the DHCP scope's DNS option (Chapter 7).

Adding a 2025 DC to an older domain: confirm the domain and forest are at the 2016 level (`(Get-ADDomain).DomainMode`, `(Get-ADForest).ForestMode`) and that SYSVOL uses DFSR (`dfsrmig /getglobalstate` should report *Eliminated*). The wizard prepares the schema automatically when run by a member of Schema Admins and Enterprise Admins.

5.6 Verify

```
Get-ADDomainController -Filter * | Format-Table Name, Site, IsGlobalCatalog, OperatingSystem
dcdiag /q                        # no output = no errors
repadmin /replsummary            # 0 fails
Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.ad.example.com
```

5.7 Troubleshooting promotion

- **"An Active Directory domain controller could not be contacted"** → the new server's DNS points to the wrong server. Point it to an existing DC and run `nltest /dsgetdc:ad.example.com`.

- **Functional level error** → raise the existing domain/forest to 2016 first.
- **SYSVOL/NETLOGON shares missing after promotion** → check DFS Replication events (Applications and Services Logs → DFS Replication); initial replication can take minutes.
- Promotion logs: `%SystemRoot%\debug\dcpromo.log` and `dcpromoui.log`.

Security considerations: DCs hold every password hash — treat them as your most sensitive servers. Run nothing else on them except DNS (and DHCP in small sites), never browse the web from them, and restrict who can sign in.

Rollback: demote a healthy DC with `Uninstall-ADDSDomainController` (or Server Manager → Remove Roles). For a failed DC, use metadata cleanup (Chapter 16).

Practical lab 5 — Build ad.example.com (Beginner, 60 min)

VMs: DC01, DC02 on an isolated switch. **Steps:** create the forest on DC01 with PowerShell; join and promote DC02 with the GUI; run the verification commands. **Expected result:** both DCs listed as Global Catalogs; `repadmin /replsummary` shows 0 failures. **Troubleshooting:** if DC02 cannot find the domain, check its DNS points to 192.0.2.10.

Knowledge check. 1) Why two DCs? *Redundancy for authentication and DNS.* 2) Minimum functional level to add a 2025 DC? *Windows Server 2016.* 3) What is DSRM for? *Offline AD repair and restore; its password is separate from domain accounts.*

6 · DNS

Overview. Active Directory depends on DNS: computers find DCs by looking up DNS SRV records. Promotion created the forward lookup zone `ad.example.com` and `_msdcs.ad.example.com`. This chapter adds reverse lookup zones, scavenging, forwarders and secure updates.

Prerequisites: a DC running DNS; Domain Admins or DnsAdmins membership.

6.1 Reverse lookup zones

A reverse zone answers "what name has this IP?" — used by troubleshooting tools, some applications and logs.

GUI: Server Manager → **Tools** → *DNS* → expand the server → right-click *Reverse Lookup Zones* → *New Zone* → *Primary zone*, tick *Store the zone in Active Directory* → replication scope *To all DNS servers running on domain controllers in this domain* → *IPv4 Reverse Lookup Zone* → Network ID `192.0.2` → *Allow only secure dynamic updates* → **Finish**.

```
Add-DnsServerPrimaryZone -NetworkId '192.0.2.0/24' -ReplicationScope Domain -DynamicUpdate
Secure
```

Create one zone per subnet range in use. If a range spans several /24 networks (e.g. 10.10.0.0–10.10.1.255), a zone on the first two octets (`10.10` → `-NetworkId 10.10.0.0/16`) covers them all. Create IPv6 reverse zones too if you use IPv6 addressing.

6.2 Secure dynamic updates

Clients register their own DNS records. *Secure only* means only authenticated domain members can create or change records — always use it for AD-integrated zones. Check: `Get-DnsServerZone | Format-Table ZoneName, DynamicUpdate`.

6.3 Aging and scavenging

Scavenging deletes stale records left by computers that left the network. It needs **both** zone aging and server scavenging enabled. Enable server scavenging on **one** DNS server only, so deletions are predictable.

GUI: DNS Manager → right-click the server → *Set Aging/Scavenging for All Zones* → tick *Scavenge stale resource records* (defaults: 7-day no-refresh, 7-day refresh) → confirm *Apply these settings to the existing Active Directory-integrated zones*. Then server *Properties* → *Advanced* → tick *Enable automatic scavenging of stale records* (period 7 days).

```
Set-DnsServerZoneAging -Name 'ad.example.com' -Aging $true -NoRefreshInterval 7.00:00:00 -
RefreshInterval 7.00:00:00
Set-DnsServerZoneAging -Name '2.0.192.in-addr.arpa' -Aging $true
Set-DnsServerScavenging -ScavengingState $true -ScavengingInterval 7.00:00:00   # on ONE
server
```

> **PRODUCTION IMPACT** Static records (servers, printers) do not age and are never scavenged. Records created dynamically for devices that are offline for longer than no-refresh + refresh (14 days by default) will be removed — set intervals longer than your DHCP lease time and school breaks if needed.

6.4 Forwarders

A **forwarder** is an upstream DNS server that your DCs ask for internet names. Configure forwarders on every DNS server.

GUI: DNS Manager → right-click the server → *Properties* → *Forwarders* → **Edit...** → add the addresses → **OK**. The checkbox *Use root hints if no forwarders are available* lets the server resolve names itself if forwarders fail.

```
Set-DnsServerForwarder -IPAddress 198.51.100.53, 198.51.100.54
Get-DnsServerForwarder
```

> **ORGANIZATION-SPECIFIC — ARKANSAS K-12 EXAMPLE FROM THE 2022 EDITION** Districts on the state network used the DIS resolvers 170.94.156.195 and 170.94.156.196 (Little Rock) and 66.204.1.66 (Fayetteville POP), and were told to remove old 165.29.x.x / 170.211.x.x entries. Districts using Cisco Umbrella content filtering used 208.67.222.222, 208.67.220.220, 208.67.222.220 and 208.67.220.222 and *unticked* root hints so that filtering could not be bypassed; Google DNS was avoided because of a testing-browser conflict. Confirm current addresses with DIS and your filtering vendor before use — this guide has not verified them. Do not point forwarders at an on-site filtering virtual appliance unless the vendor documents that design.

6.5 Conditional forwarders

Send queries for a specific domain (a partner district, a vendor VPN zone) to specific servers: right-click *Conditional Forwarders* → *New Conditional Forwarder*, or

```
Add-DnsServerConditionalForwarderZone -Name 'partner.example.org' -MasterServers 203.0.113.10
-ReplicationScope Forest
```

6.6 Verify and troubleshoot

```
Resolve-DnsName dc01.ad.example.com
Resolve-DnsName 192.0.2.10                       # reverse lookup
Resolve-DnsName www.microsoft.com -Server 192.0.2.10
dcdiag /test:dns /v
Clear-DnsServerCache; Clear-DnsClientCache
```

Decision path: client cannot resolve → is the client's DNS server a DC? (`ipconfig /all`) → can it reach the DC on port 53? (`Test-NetConnection 192.0.2.10 -Port 53`) → internal names fail = check the zone and SRV records; internet names fail = check forwarders and outbound firewall → still failing = DNS Server event log.

Security: keep zones AD-integrated with secure updates; limit membership of *DnsAdmins* (members can effectively gain control of DCs); DNS policies, DNSSEC and response rate limiting are covered in the expansion plan (Appendix I).

7 · DHCP

Overview. DHCP hands out IP addresses and settings (gateway, DNS servers, domain name) to clients automatically. In Active Directory, a Windows DHCP server must be **authorized** before it will hand out addresses — this stops rogue Windows DHCP servers.

Prerequisites: a domain-joined server with a static IP (a DC is acceptable in small sites); Enterprise Admins rights to authorize; the subnet, range, exclusions and options for each scope.

7.1 Install and authorize

GUI: Add Roles and Features → **DHCP Server** → **Add Features** → **Install**. Then the notifications flag → *Complete DHCP configuration* → **Next** → *Authorization* with your admin credentials → **Commit**. This also creates the local *DHCP Administrators* and *DHCP Users* groups.

```
Install-WindowsFeature DHCP -IncludeManagementTools
netsh dhcp add securitygroups
Restart-Service DHCPServer
Add-DhcpServerInDC -DnsName 'dc01.ad.example.com' -IPAddress 192.0.2.10
Set-ItemProperty 'HKLM:\SOFTWARE\Microsoft\ServerManager\Roles\12' -Name ConfigurationState -
Value 2   # clears the Server Manager reminder
```

7.2 Create a scope

GUI: Tools → *DHCP* → expand the server → right-click *IPv4* → *New Scope*: name (e.g. *HS Student VLAN 20*) → start/end address and subnet mask → exclusions → lease duration → *Yes, I want to configure these options now*: router, DNS servers and domain name → skip WINS → *Activate*.

```
Add-DhcpServerv4Scope -Name 'Lab clients' -StartRange 192.0.2.100 -EndRange 192.0.2.200 `
  -SubnetMask 255.255.255.0 -LeaseDuration 1.00:00:00 -State Active
Set-DhcpServerv4OptionValue -ScopeId 192.0.2.0 -Router 192.0.2.1 `
  -DnsServer 192.0.2.10,192.0.2.11 -DnsDomain 'ad.example.com'
Add-DhcpServerv4ExclusionRange -ScopeId 192.0.2.0 -StartRange 192.0.2.100 -EndRange
192.0.2.109
Add-DhcpServerv4Reservation -ScopeId 192.0.2.0 -IPAddress 192.0.2.150 -ClientId 'AA-BB-CC-DD-
EE-FF' -Description 'Library printer'
```

Addressing tip: keep the start of each subnet outside the scope for statically addressed devices (servers, switches, printers), as the 2022 edition suggested — or use reservations so every address is still documented in DHCP.

Common options

OPTION	NAME	VALUE
003	Router	The subnet's default gateway
006	DNS Servers	Your DCs only
015	DNS Domain Name	`ad.example.com`
044 / 046	WINS servers / node type	**Do not set** unless a documented legacy application still needs WINS (Appendix D)

Set options at *Server Options* when they are the same for every scope (DNS servers, domain name) and at *Scope Options* when they differ (router).

7.3 DNS registration from DHCP

When DHCP runs on a DC, give it a dedicated low-privilege account for DNS updates; otherwise it registers records using the DC's own computer account, which is a security risk.

```
New-ADUser -Name 'svc-dhcpdns' -AccountPassword (Read-Host -AsSecureString) -Enabled $true -
PasswordNeverExpires $true
Set-DhcpServerDnsCredential -Credential (Get-Credential 'AD\svc-dhcpdns')
Set-DhcpServerv4DnsSetting -DynamicUpdates OnClientRequest -DeleteDnsRROnLeaseExpiry $true
```

7.4 Bindings and conflict detection

- **Bindings:** right-click *IPv4* → *Properties* → *Advanced* → **Bindings…**. Only the adapter on the served network should be ticked.
- **Conflict detection** makes the server ping an address before leasing it. Each attempt adds about a second per lease. The 2022 edition set 3 attempts; use 0 (default) with failover, or 1–2 only while troubleshooting duplicate-address problems.

7.5 DHCP failover (high availability)

Two DHCP servers share a scope so clients keep getting addresses when one is down. **Load balance** mode splits clients (e.g. 50/50) — best for two servers in the same site. **Hot standby** keeps one server passive with a small reserved percentage — best for a central backup server for many sites.

GUI: right-click the scope → *Configure Failover…* → choose partner server → mode → set a shared secret → **Finish**.

```
Install-WindowsFeature DHCP -IncludeManagementTools -ComputerName DC02
Add-DhcpServerInDC -DnsName 'dc02.ad.example.com' -IPAddress 192.0.2.11
Add-DhcpServerv4Failover -ComputerName 'dc01.ad.example.com' -PartnerServer
'dc02.ad.example.com' `
  -Name 'DC01-DC02' -ScopeId 192.0.2.0 -LoadBalancePercent 50 -MaxClientLeadTime 01:00:00 `
  -SharedSecret '<long-random-secret>'
Get-DhcpServerv4Failover
```

After changing scope options on one server, replicate them: right-click the scope → *Replicate Scope*, or `Invoke-DhcpServerv4FailoverReplication -Name 'DC01-DC02'`. Failover partners communicate on TCP 647.

7.6 Back up, export and restore DHCP

Windows backs up the DHCP database every 60 minutes to `%SystemRoot%\System32\dhcp\backup`. Copy it off the server, and keep an export for migrations.

```
# Export configuration and leases (use for migration to a new server)
Export-DhcpServer -ComputerName dc01 -File C:\DHCPExport\dhcp.xml -Leases -Force
# Import on the new, authorized server
Import-DhcpServer -ComputerName srv02 -File C:\DHCPExport\dhcp.xml -BackupPath C:\DHCPBackup -
Leases
# Point-in-time backup / restore of the database
Backup-DhcpServer -ComputerName dc01 -Path C:\DHCPBackup
Restore-DhcpServer -ComputerName dc01 -Path C:\DHCPBackup     # then Restart-Service
DHCPServer
```

The 2022 edition's `netsh dhcp server export / import` commands still work but are a legacy method; the PowerShell cmdlets above are preferred.

Verify: on a client run `ipconfig /release` then `ipconfig /renew` and `ipconfig /all` — the address, gateway, DNS servers and suffix should match the scope. On the server: `Get-DhcpServerv4ScopeStatistics`.

Troubleshooting: *Clients get 169.254.x.x* → server not authorized, scope inactive or full, or the router lacks a DHCP relay (IP helper) for that VLAN. *Red down-arrow on the server* → not authorized. Logs: `%SystemRoot%\System32\dhcp\DhcpSrvLog-*.log`.

8 · Time synchronization

Overview. Domain members take time from DCs; DCs take time from the DC holding the **PDC emulator** role; only the PDC emulator should take time from an external source. Kerberos tolerates 5 minutes of difference.

1. Find the PDC emulator: `netdom query fsmo` or `(Get-ADDomain).PDCEmulator`.
2. On the PDC emulator (elevated prompt):

```
w32tm /config /manualpeerlist:"time.windows.com,0x8 time.nist.gov,0x8"
/syncfromflags:manual /reliable:yes /update
Restart-Service w32time
w32tm /resync /rediscover
```

3. On every other DC (in case they were configured manually in the past):

```
w32tm /config /syncfromflags:domhier /update
Restart-Service w32time
```

Verify: `w32tm /query /status` and `w32tm /query /source` on the PDC show the external peer; on other machines the source is a DC. `w32tm /monitor` shows offsets for all DCs.

> **ORGANIZATION-SPECIFIC** The 2022 edition used state time servers (`165.29.1.11`, `170.94.1.1` and hostnames in the `state.ar.us` domain, one of which was misspelled). Because the same edition told readers to remove 165.29.x.x DNS servers, confirm with DIS which time sources are current before using them. Public alternatives: `time.windows.com`, `time.nist.gov`, `us.pool.ntp.org`. Allow outbound UDP 123 from the PDC emulator only.

Virtualized DCs: make sure the PDC emulator does not also take time from the hypervisor host; follow your hypervisor vendor's guidance for domain controllers. If you move the PDC emulator role, repeat step 2 on the new holder and step 3 on the old one.

9 · Windows updates & WSUS

Overview. Windows Server Update Services (WSUS) downloads Microsoft updates once and distributes them to your computers on your schedule. In Windows Server 2025 WSUS is **deprecated but fully supported**: it is included and receives update content, but gets no new features. Plan for an eventual replacement (Section 9.7) while using it safely.

Prerequisites: a member server (not a DC) with a dedicated data volume — allow 200 GB+ for content (much less if you only sync the products you use); WSUS installs IIS 10 and uses Windows Internal Database (WID) by default, or SQL Server for large deployments.

> **CORRECTION** The 2022 edition listed IIS 6.0, .NET Framework 2.0 and Windows 8 as requirements. These are long obsolete; Server Manager installs the correct IIS and .NET components automatically.

9.1 Install

GUI: Add Roles and Features → **Windows Server Update Services** → **Add Features** → role services: *WID Connectivity* + *WSUS Services* (or *SQL Server Connectivity*) → content location `D:\WSUS` → accept the IIS defaults → **Install**. Then **Tools** → *Windows Server Update Services* → **Run** post-installation tasks.

```
Install-WindowsFeature UpdateServices -IncludeManagementTools
& 'C:\Program Files\Update Services\Tools\wsusutil.exe' postinstall CONTENT_DIR=D:\WSUS
```

9.2 Configuration wizard

1. **Upstream server:** *Synchronize from Microsoft Update* (or from your central WSUS server for a downstream site).
2. **Proxy:** only if your network requires one.
3. **Start Connecting** — downloads the product catalogue; this can take a while.
4. **Languages:** only those you use (e.g. English).
5. **Products:** only what you run — e.g. *Windows Server 2025*, *Windows 11*, *Microsoft Defender Antivirus*, Office/Microsoft 365 Apps if managed by WSUS. Every extra product adds storage and database load.
6. **Classifications:** Critical Updates, Security Updates, Definition Updates, Update Rollups, Updates. Add *Upgrades* only if you deploy feature updates through WSUS. Leave *Drivers* off (driver sync is deprecated and bloats the database).
7. **Schedule:** synchronize automatically, off-peak (e.g. 23:00).

9.3 Staged approvals instead of “approve everything”

> **CORRECTION** The 2022 edition auto-approved every classification for all computers. One bad update can then reach every machine at once. Use deployment rings instead.

1. In the WSUS console: *Options* → *Computers* → *Use Group Policy or registry settings on computers* (client-side targeting).
2. Create computer groups: `Ring0-IT`, `Ring1-Pilot`, `Ring2-Workstations`, `Ring3-Servers`.
3. Automatic approval: keep only *Definition Updates* auto-approved for all computers (Defender signatures change several times a day).
4. Each Patch Tuesday (second Tuesday): approve security and quality updates for Ring0 the same day → Ring1 after 2–3 days without problems → Ring2 after about a week → servers in their maintenance window.
5. Monthly: run *Options* → *Server Cleanup Wizard* and decline superseded updates, so the database stays healthy.

```
Get-WsusServer | Get-WsusUpdate -Classification Security -Approval Unapproved -Status
FailedOrNeeded |
  Approve-WsusUpdate -Action Install -TargetGroupName 'Ring0-IT'
Invoke-WsusServerCleanup -DeclineSupersededUpdates -CleanupObsoleteUpdates -
CleanupUnneededContentFiles
```

9.4 Group Policy for clients

Create separate GPOs for workstations and servers so servers never restart unexpectedly. In GPMC create `WSUS - Workstations`, then edit: Computer Configuration → Policies → Administrative Templates → Windows Components → *Windows Update*. In current templates the settings are grouped into sub-folders:

SETTING (FOLDER)	VALUE
Specify intranet Microsoft update service location (Manage updates offered from Windows Server Update Service)	Enabled; both boxes `https://srv01.ad.example.com:8531` (or `http://…:8530` until HTTPS is configured)
Enable client-side targeting (same folder)	Enabled; group name `Ring2-Workstations`
Configure Automatic Updates (Manage end user experience)	Enabled; option 4 — Auto download and schedule the install; choose day/time
Automatic Updates detection frequency	Enabled; default 22 hours is fine
No auto-restart with logged on users for scheduled automatic updates installations	Enabled for workstations where unsaved work matters

Link it to the workstations OU. Create `WSUS - Servers` with option 3 (download and notify) or a maintenance-window schedule, targeting `Ring3-Servers`, and link it to the servers OU.

> **SECURITY RISK** Plain HTTP (8530) lets an attacker on your network tamper with update metadata. Configure WSUS with an SSL certificate (port 8531) — see Microsoft Learn, *Secure WSUS with SSL*.

9.5 Verify

```
# On a client
gpupdate /force
Get-ItemProperty 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate'
# Settings > Windows Update > Check for updates — should report "managed by your organization"
```

The client appears in its computer group in the WSUS console after its first report (can take a few hours; it does not appear in the group if client-side targeting is off).

9.6 Troubleshooting

- **Client never appears** → wrong URL/port in GPO, firewall blocks 8530/8531, or the computer is a clone with a duplicate WSUS ID.
- **Console times out / crashes** → database bloat; run the cleanup wizard and re-index the WID database.
- **Update fails on client** → `Get-WindowsUpdateLog` creates a readable log on the desktop.

9.7 Modern alternatives

Azure Update Manager (servers, through Azure Arc), Microsoft Intune with Windows Autopatch (Windows clients), Microsoft Configuration Manager (uses WSUS components internally), and hotpatching for eligible Windows Server 2025 servers. Choose based on licensing, internet connectivity and the number of devices.

10 · AD structure, users & groups (K–12 example)

Overview. Organizational units (OUs) are folders for objects; you link Group Policy to them and delegate administration through them. Security groups grant access to resources. A clear structure makes everything later — policies, permissions, LAPS — simpler.

Prerequisites: a working domain; Domain Admins (or delegated rights) and RSAT tools (Active Directory Users and Computers, ADUC; Active Directory Administrative Center, ADAC).

10.1 Recommended OU design

```
ad.example.com
├── Domain Controllers            (default — leave as is)
├── Admin                         (tier-0 admin accounts and groups; tightly restricted)
└── District
    ├── Groups                    (security groups: Faculty, Students, role groups)
    ├── Servers
    ├── Service Accounts
    ├── Users
    │   ├── Faculty
    │   └── Students
    │       ├── 2030              (graduation year, not grade level)
    │       └── 2031
    └── Workstations
        ├── Faculty
        ├── Labs
        └── Student
```

Use one OU per graduation year so a cohort moves through school without moving objects. For multiple sites, add a site level under *Users* and *Workstations* only if different people administer each site; otherwise keep it flat and use groups.

New computers and users land in the default *Computers* and *Users* containers, where you cannot link GPOs. Redirect them:

```
redircmp "OU=Workstations,OU=District,DC=ad,DC=example,DC=com"
redirusr "OU=Users,OU=District,DC=ad,DC=example,DC=com"
```

10.2 Create OUs and groups

GUI: ADUC → right-click the domain → *New* → *Organizational Unit* → name → keep *Protect container from accidental deletion* ticked. For groups: right-click *Groups* OU → *New* → *Group* → scope *Global*, type *Security*.

```
$base = 'DC=ad,DC=example,DC=com'
New-ADOrganizationalUnit -Name 'District' -Path $base
'Groups','Servers','Service Accounts','Users','Workstations' |
  ForEach-Object { New-ADOrganizationalUnit -Name $_ -Path "OU=District,$base" }
'Faculty','Students' | ForEach-Object { New-ADOrganizationalUnit -Name $_ -Path
"OU=Users,OU=District,$base" }
New-ADGroup -Name 'Faculty'  -GroupScope Global -GroupCategory Security -Path
"OU=Groups,OU=District,$base"
New-ADGroup -Name 'Students' -GroupScope Global -GroupCategory Security -Path
"OU=Groups,OU=District,$base"
```

Group strategy (AGDLP): put **A**ccounts into **G**lobal role groups (Faculty, Yearbook), put those into **D**omain **L**ocal resource groups (`FS01-Yearbook-Modify`), and grant **P**ermissions only to the domain local group. Never grant permissions to individual users.

10.3 Separate administrator accounts

Each administrator gets a normal account for email and browsing (`jsmith`) and a separate admin account (`admin.jsmith`) used only for administration. Add admin accounts to the narrowest group that works — *Domain Admins* only for those who manage DCs and the directory itself.

10.4 User template (GUI method)

Templates still save time for one-off accounts.

1. ADUC → *Faculty* OU → *New* → *User*: first name `_Faculty`, last name `Template`, logon `_facultytemplate` (the underscore keeps it at the top of the list).
2. Set a long random password, tick *User must change password at next logon* and ***Account is disabled*** — templates must always stay disabled.
3. Properties → *Member Of* → add *Faculty*. *Profile* → *Home folder* → *Connect* `H:` to `\\srv01\Faculty-Homes$\%username%`. Leave *Logon script* empty (use Group Policy, Section 10.6).
4. To create a user: right-click the template → *Copy…* → enter the person's details → untick *Account is disabled*. ADUC creates the home folder with the right permissions.

10.5 Bulk import with PowerShell (replaces the Excel/dsadd batch file)

The 2022 edition built `dsadd` commands in Excel with the same default password (`Password1$`) for every account. That password was printed in the manual and works for any account that has not yet signed in. The script below gives every account a unique random temporary password, checks for duplicates, creates home folders and supports a dry run.

Purpose: create faculty and student accounts from a CSV export (e.g. from the student information system). **Prerequisites:** ActiveDirectory module (RSAT), the OU structure in 10.1, the home folder shares in Chapter 11. **Permissions:** create users in the target OUs and write to the home folder roots. **Environment:** Windows PowerShell 5.1 or PowerShell 7 on Windows Server 2025 or a Windows admin workstation.

CSV columns: `FirstName,LastName,Role,GradYear` (Role = Student or Faculty; GradYear blank for faculty).

```
#Requires -Modules ActiveDirectory
[CmdletBinding(SupportsShouldProcess)]
param(
  [Parameter(Mandatory)][string]$CsvPath,
  [string]$OutputPath = '.\new-accounts.csv',
  [string]$UpnSuffix  = 'ad.example.com',
  [string]$Domain     = 'AD',
  [string]$BaseDN     = 'OU=District,DC=ad,DC=example,DC=com',
  [string]$FileServer = 'srv01'
)
$ErrorActionPreference = 'Stop'
$ti  = (Get-Culture).TextInfo
$rng = [System.Security.Cryptography.RandomNumberGenerator]::Create()
function New-TempPassword {
  $bytes = New-Object byte[] 15; $rng.GetBytes($bytes)
  ([Convert]::ToBase64String($bytes) -replace '[+/=]', 'x') + '#7a'   # 20+ chars, all four
character types
}
$results = foreach ($row in Import-Csv -Path $CsvPath) {
  $sam = "$($row.FirstName).$($row.LastName)"
  try {
    $first = $ti.ToTitleCase($row.FirstName.Trim().ToLower())
    $last  = $ti.ToTitleCase($row.LastName.Trim().ToLower())
    $sam   = ("$first.$last" -replace "[^A-Za-z0-9.\-]", '')
    if ($sam.Length -gt 20) { $sam = $sam.Substring(0, 20) }
    switch ($row.Role) {
      'Student' { $ou = "OU=$($row.GradYear),OU=Students,OU=Users,$BaseDN"; $group =
'Students'; $share = 'Student-Homes$' }
      'Faculty' { $ou = "OU=Faculty,OU=Users,$BaseDN";                        $group =
'Faculty';  $share = 'Faculty-Homes$' }
      default   { throw "Unknown role '$($row.Role)'" }
    }
    if (Get-ADUser -Filter "SamAccountName -eq '$sam'") { throw "$sam already exists" }
    $homePath = "\\$FileServer\$share\$sam"   # ($home is a reserved PowerShell variable)
    $pw   = New-TempPassword
    if ($PSCmdlet.ShouldProcess($sam, 'Create user')) {
      New-ADUser -Name "$first $last" -GivenName $first -Surname $last -SamAccountName $sam `
        -UserPrincipalName "$sam@$UpnSuffix" -Path $ou -Enabled $true -ChangePasswordAtLogon
$true `
        -AccountPassword (ConvertTo-SecureString $pw -AsPlainText -Force) -HomeDrive 'H:' -
HomeDirectory $homePath
      Add-ADGroupMember -Identity $group -Members $sam
      New-Item -ItemType Directory -Path $homePath -Force | Out-Null
      icacls $homePath /grant "$Domain\${sam}:(OI)(CI)M" | Out-Null
    }
    [pscustomobject]@{ SamAccountName = $sam; TempPassword = $pw; Status = 'Created' }
  } catch {
```

```
    [pscustomobject]@{ SamAccountName = $sam; TempPassword = ''; Status = "Skipped:
$($_.Exception.Message)" }
  }
}
$results | Export-Csv -Path $OutputPath -NoTypeInformation
$results | Group-Object Status | Format-Table Count, Name
```

Usage: dry run first — `.\New-BulkUsers.ps1 -CsvPath .\students.csv -WhatIf` — then run without `-WhatIf`. **Expected output:** a count of Created/Skipped rows and `new-accounts.csv`. **Verify:** `Get-ADUser -Filter * -SearchBase "OU=2031,OU=Students,OU=Users,OU=District,DC=ad,DC=example,DC=com" | Measure-Object`.

> **SECURITY RISK** The output file contains temporary passwords. Store it on an encrypted, access-restricted location, deliver passwords through a secure channel, and delete the file once distributed. Students who never sign in keep a valid temporary password — consider creating accounts disabled and enabling them when the student starts.

Validation status: documentation-verified syntax; not yet executed in a lab.

10.6 Drive mappings — Group Policy Preferences instead of logon.bat

The 2022 edition used a `logon.bat` with `NET USE`, an `Xcopy` of desktop icons and a VBScript to rename drives. VBScript is deprecated and will be removed from Windows; Group Policy Preferences do all three jobs without scripts, apply per group, and label the drive directly.

1. GPMC → create GPO `User - Drive Maps` → edit → User Configuration → Preferences → Windows Settings → *Drive Maps* → right-click → *New* → *Mapped Drive*.
2. Action **Update**; Location `\\srv01\Apps`; *Label as* `Apps`; Drive letter `N:`; tick *Reconnect*.
3. *Common* tab → tick *Run in logged-on user's security context* and *Item-level targeting* → *Targeting…* → *New Item* → *Security Group* → `AD\Faculty`.
4. Repeat for each drive. Desktop shortcuts: User Configuration → Preferences → Windows Settings → *Shortcuts*.
5. Link the GPO to `OU=Users,OU=District`.

Home drives (H:) come from the account's profile setting (10.4). BGInfo (Sysinternals) can still be launched from a GPO logon script if you use it. Existing `NETLOGON` batch files keep working; migrate them before VBScript removal affects any `.vbs` helpers.

11 · File services: shares, home folders, shadow copies, quotas

Overview. Access to a shared folder is limited by **both** share permissions and NTFS permissions — the most restrictive wins. A simple, reliable pattern is: share permission *Authenticated Users: Full Control* (or Change), and do the real access control with NTFS permissions on the folder.

Prerequisites: a member file server (SRV01) with a separate data volume (e.g. `D:`, ideally separate volumes for faculty and student home folders, as the 2022 edition recommended).

11.1 Home folder root (Faculty-Homes$)

The `$` at the end hides the share from browsing. Microsoft's recommended NTFS permissions for a home folder root let users create only their own folder:

PRINCIPAL	NTFS PERMISSION	APPLIES TO
SYSTEM	Full control	This folder, subfolders and files
Administrators	Full control	This folder, subfolders and files
CREATOR OWNER	Full control	Subfolders and files only
Faculty (group)	List folder / read data, Create folders / append data, Read attributes	This folder only

GUI: create `D:\Faculty-Homes` → Properties → *Sharing* → *Advanced Sharing* → share name `Faculty-Homes$` → *Permissions*: remove *Everyone*, add *Authenticated Users* Change and *Administrators* Full Control → *Caching*: *No files or programs... available offline*. Then *Security* → *Advanced* → *Disable inheritance* → *Convert inherited permissions* → remove *Users* entries and set the table above.

```
New-Item -ItemType Directory D:\Faculty-Homes | Out-Null
icacls D:\Faculty-Homes /inheritance:r /grant:r 'SYSTEM:(OI)(CI)F' 'BUILTIN\Administrators:
(OI)(CI)F' `
  'CREATOR OWNER:(OI)(CI)(IO)F' 'AD\Faculty:(RD,AD,RA)'
New-SmbShare -Name 'Faculty-Homes$' -Path D:\Faculty-Homes -FullAccess
'BUILTIN\Administrators' `
  -ChangeAccess 'NT AUTHORITY\Authenticated Users' -CachingMode None -FolderEnumerationMode
AccessBased
```

Access-based enumeration (`-FolderEnumerationMode AccessBased`) hides folders a user cannot open, so students see only their own folder. Repeat for `Student-Homes$` with the *Students* group.

Verify: sign in as a test faculty user — `H:` maps, they can create files, and cannot open another user's folder. `Get-SmbShare`, `Get-Acl D:\Faculty-Homes | Format-List`.

11.2 Shadow Copies of Shared Folders (Previous Versions)

Shadow copies take scheduled point-in-time snapshots of a volume so users can restore an earlier version of a file themselves: right-click the file or folder → *Properties* → *Previous Versions*. Only changed blocks are stored, so space use is modest.

> **NOTE** Shadow copies are **not backups**. They live on the same server, a maximum of 64 are kept per volume, older ones are deleted when space runs out, and ransomware often deletes them. Keep a separate backup (Chapter 17).

1. File Explorer → *This PC* → right-click the data volume → *Configure Shadow Copies…*
2. Select the volume → **Settings…** → set storage limit (10–20% of the volume; on busy volumes store on another volume) → **Schedule…**. The default is 7:00 AM and 12:00 PM, Monday–Friday; add times that suit the school day.
3. **Enable** → **Create Now** to take the first copy.

```
vssadmin add shadowstorage /for=D: /on=D: /maxsize=15%
vssadmin create shadow /for=D:
vssadmin list shadows /for=D:
```

11.3 Quotas

NTFS volume quotas (volume Properties → *Quota*) apply one limit to every user on a whole volume, based on file ownership. **File Server Resource Manager (FSRM)** quotas apply to folders and are more flexible — preferred for home folders.

Install FSRM (no restart needed):

```
Install-WindowsFeature FS-Resource-Manager -IncludeManagementTools
```

GUI: Tools → *File Server Resource Manager* → *Quota Management* → *Quota Templates* → *Create Quota Template…* (name, limit, *Hard quota* stops saving, *Soft quota* only warns; add email thresholds). Then *Quotas* → *Create Quota…* → path `D:\Student-Homes` → *Auto apply template and create quotas on existing and new subfolders* → choose the template.

```
New-FsrmQuotaTemplate -Name 'Student Home 5 GB' -Size 5GB
New-FsrmQuotaTemplate -Name 'Faculty Home 20 GB' -Size 20GB
New-FsrmAutoQuota -Path 'D:\Student-Homes' -Template 'Student Home 5 GB'
New-FsrmAutoQuota -Path 'D:\Faculty-Homes' -Template 'Faculty Home 20 GB'
New-FsrmQuota -Path 'D:\Shared\Yearbook' -Template 'Faculty Home 20 GB'    # one shared limit
for a group folder
```

File screening (optional) blocks file types, e.g. executables in student home folders: *File Screening Management* → *Create File Screen* → path → file group *Executable Files* → *Active screening*.

Verify: `Get-FsrmQuota -Path 'D:\Student-Homes\*'` lists per-user quotas with usage.

12 · Password policies & fine-grained password policies

Overview. The domain password policy lives in the *Default Domain Policy* GPO and applies to every domain account. **Fine-grained password policies (FGPP)**, or Password Settings Objects (PSOs), let you apply different rules to specific groups — for example staff vs students.

> **CORRECTION** The 2022 edition said FGPP was introduced in Windows Server 2022. FGPP has existed since Windows Server 2008, with a graphical editor in ADAC since 2012. It also gave both example policies precedence 1; give every PSO a unique precedence (lower number wins).

12.1 Choosing values

Your state or district policy may set required values — the 2022 edition followed Arkansas **ACT 723** K–12 security policy. Treat the table below as the organization's example and check the current text of your own policy. For reference, NIST SP 800-63B (Revision 4) recommends long passwords (at least 15 characters when a password is the only factor), screening against known-breached passwords, and *not* forcing periodic changes unless compromise is suspected. Where your mandated policy differs, follow the mandate and document the difference.

SETTING	FACULTY/STAFF	STUDENTS
Precedence	10	20
Applies to group	Faculty	Students
Minimum length	8 (ACT 723 example) — consider 12–15	8
Enforce password history	5 (recommended in original) — consider 24	5
Minimum password age (days)	1	1
Maximum password age (days)	90	180
Complexity	Enabled	Enabled
Lockout threshold / duration	10 attempts / 15 min	10 attempts / 15 min

Create a third, stricter PSO (e.g. 20+ characters) for administrative and service accounts, precedence 1.

12.2 Create a PSO

GUI: Tools → *Active Directory Administrative Center* → switch to *Tree View* → domain → *System* → *Password Settings Container* → right-click → *New* → *Password Settings* → enter values → under *Directly Applies To* click **Add…** → *Faculty* → **OK**.

```
New-ADFineGrainedPasswordPolicy -Name 'PSO-Faculty' -Precedence 10 -MinPasswordLength 8 `
  -PasswordHistoryCount 5 -MinPasswordAge 1.00:00:00 -MaxPasswordAge 90.00:00:00 `
  -ComplexityEnabled $true -ReversibleEncryptionEnabled $false `
  -LockoutThreshold 10 -LockoutDuration 00:15:00 -LockoutObservationWindow 00:15:00
Add-ADFineGrainedPasswordPolicySubject -Identity 'PSO-Faculty' -Subjects 'Faculty'
```

Verify: `Get-ADUserResultantPasswordPolicy -Identity jane.smith` shows which PSO wins for a user. PSOs apply to users and *global* security groups only — not OUs.

13 · Group Policy: common policies & troubleshooting

Overview. A Group Policy Object (GPO) is a set of settings linked to a site, domain or OU. Computer settings apply at startup and every ~90 minutes; user settings at sign-in and every ~90 minutes.

> **CORRECTION** The 2022 edition placed most settings in the *Default Domain Policy*. Keep that GPO for the domain password, lockout and Kerberos policy only. Create a separate, clearly named GPO for each purpose, so you can test, delegate, back up and roll back each one on its own.

Before you start: create the **Central Store** so all admins see the same, current Administrative Templates. Download the latest Windows 11/Server ADMX templates from Microsoft and copy the `PolicyDefinitions` folder to `\\ad.example.com\SYSVOL\ad.example.com\Policies\PolicyDefinitions`.

Basic steps used below: Tools → *Group Policy Management* → *Forest* → *Domains* → *ad.example.com* → right-click *Group Policy Objects* → *New* → name → right-click the GPO → *Edit*. Link it by dragging it onto an OU (or right-click OU → *Link an Existing GPO*).

13.1 Advanced audit policy (replaces the basic Audit Policy)

GPO `Computer - Audit Policy`, linked to the domain (or to *Domain Controllers* and server OUs). Path: Computer Configuration → Policies → Windows Settings → Security Settings → *Advanced Audit Policy Configuration* → Audit Policies. Also enable Security Options → *Audit: Force audit policy subcategory settings… to override audit policy category settings*.

CATEGORY → SUBCATEGORY	AUDIT
Account Logon → Credential Validation; Kerberos Authentication Service; Kerberos Service Ticket Operations	Success, Failure
Account Management → User Account Management; Security Group Management; Computer Account Management	Success (+Failure for users)
Logon/Logoff → Logon; Account Lockout; Special Logon	Success, Failure / Failure / Success
Policy Change → Audit Policy Change; Authentication Policy Change	Success
DS Access → Directory Service Changes (DCs)	Success
Detailed Tracking → Process Creation	Success

The Windows Server 2025 security baseline (Chapter 15) sets a fuller list; if you apply it, do not duplicate these settings.

13.2 Security event log size and retention

The 2022 edition set *Retain security log* to 90 days. That option stops *overwriting* events newer than 90 days, so when the log fills, new events can be lost. Size the log instead and forward events to a central collector or SIEM for long-term retention.

- Computer Configuration → Policies → Administrative Templates → Windows Components → Event Log Service → Security → *Specify the maximum log file size (KB)*: 1,048,576 (1 GB) on DCs, 262,144+ on other servers.
- *Back up log automatically when full* creates archive files — only enable it if disk space is monitored, and copy archives elsewhere.
- For 90-day retention, use Windows Event Forwarding or your SIEM.

13.3 Logon banner

GPO `Computer - Logon Banner`: Security Settings → Local Policies → Security Options → *Interactive logon: Message title for users attempting to log on* and *Message text…*. Have the text approved by your district's legal or policy office.

13.4 Screen lock

- Computer policy (applies to everyone on the machine): Security Options → *Interactive logon: Machine inactivity limit* = 900 seconds.
- User policy (original method, still valid): User Configuration → Policies → Administrative Templates → Control Panel → Personalization → *Enable screen saver*, *Password protect the screen saver*, *Screen saver timeout* = 900.

13.5 Folder redirection

GPO `User - Folder Redirection`: User Configuration → Policies → Windows Settings → Folder Redirection → right-click *Documents* → *Properties* → *Basic – Redirect everyone's folder to the same location* → *Redirect to the user's home directory*. On the *Settings* tab, leave the legacy *Also apply redirection policy to Windows 2000…* option **unticked** (obsolete). When prompted, choose whether to redirect Pictures, Music and Videos too. If your district uses OneDrive, *Known Folder Move* is the modern alternative.

13.6 Restrict computers to faculty use only

1. Create a computer security group `Faculty Use Only Computers` and add the computer accounts.
2. Create GPO `Computer - Faculty Use Only`. On its *Scope* tab, remove *Authenticated Users* from Security Filtering and add `Faculty Use Only Computers`. On the *Delegation* tab add *Authenticated Users* with *Read* only (required since 2016 for GPOs to be readable).
3. Edit: Security Settings → Local Policies → User Rights Assignment → *Allow log on locally* → Define: `Administrators`, `AD\Faculty`. This *replaces* the default list, so include every group that must sign in.
4. Link to the workstations OU containing these computers.

Computers pick up a new group membership after a restart. To refresh without restarting: `klist -li 0x3e7 purge` then `gpupdate /force`.

13.7 Apply and check policy

```
gpupdate                          # refresh changed policy
gpupdate /target:computer /force  # reapply all computer settings
gpupdate /force /boot             # restart if a setting needs it
gpresult /r                       # summary for the current user and computer
gpresult /scope computer /r       # computer part (run elevated)
gpresult /h C:\Temp\gp.html /f    # full HTML report — the best troubleshooting view
Invoke-GPUpdate -Computer CLIENT01 -RandomDelayInMinutes 0 -Force
```

GPMC can also push a refresh to every computer in an OU: right-click the OU → *Group Policy Update…* (available since Windows Server 2012; needs the remote scheduled task and WMI firewall rules on clients). *Resultant Set of Policy* (`rsop.msc`) still works but does not show every modern setting; prefer `gpresult /h`. In GPMC, *Group Policy Results* and *Group Policy Modeling* wizards give the same view remotely.

13.8 Troubleshooting decision path

1. Is the object (user or computer) in an OU where the GPO is linked? Is the link enabled?
2. Does Security Filtering include the user/computer (or a group it belongs to), and do *Authenticated Users* or *Domain Computers* have Read?
3. Is a WMI filter excluding it? Is inheritance blocked on the OU, or a conflicting GPO winning (higher link order)?
4. Run `gpresult /h` — look at *Denied GPOs* and the reason.
5. Check Event Viewer → Applications and Services Logs → Microsoft → Windows → GroupPolicy → Operational, and that the client can reach SYSVOL: `dir \\ad.example.com\SYSVOL`.

Back up GPOs before editing: `Backup-GPO -All -Path D:\GPOBackups`; restore with `Restore-GPO -Name '<GPO>' -Path D:\GPOBackups`.

14 · Windows LAPS

Overview. If every computer has the same local Administrator password, stealing it from one machine unlocks them all. **Windows LAPS** (Local Administrator Password Solution) gives every computer a unique, random local admin password, rotates it automatically and stores it securely in Active Directory (or Microsoft Entra ID). It is built into Windows Server 2025, Windows Server 2019/2022 and Windows 10/11 with current updates — nothing to install.

You will learn to extend the schema, grant permissions, configure policy, retrieve passwords and protect DC recovery (DSRM) passwords.

Prerequisites: Schema Admins (schema update, once per forest); Domain Admins for permissions and GPO; domain functional level 2016 for password encryption; clients on supported Windows versions.

14.1 How it works

Group Policy tells each computer which account to manage, how complex and how old the password may be, and where to back it up. The computer itself generates the password, writes it — encrypted — to its own computer object attribute (`msLAPS-EncryptedPassword`), then sets it locally. Only principals you authorize can decrypt and read it. After someone uses the password, a *post-authentication action* can rotate it automatically.

Windows LAPS vs legacy Microsoft LAPS: the legacy product was a separate MSI that stored plaintext passwords in `ms-Mcs-AdmPwd`. It is deprecated and not supported on the newest Windows releases. Windows LAPS uses new attributes, encrypts by default, supports Entra ID and DSRM, and has PowerShell built in. It can run side by side during migration as long as both don't manage the same account.

14.2 Prepare Active Directory

```
# 1. Extend the schema (Schema Admins, once per forest)
Update-LapsADSchema -Verbose

# 2. Let computers in an OU write their own password
Set-LapsADComputerSelfPermission -Identity
'OU=Workstations,OU=District,DC=ad,DC=example,DC=com'
Set-LapsADComputerSelfPermission -Identity 'OU=Servers,OU=District,DC=ad,DC=example,DC=com'

# 3. Create a reader group and allow it to read passwords
New-ADGroup -Name 'LAPS Readers - Workstations' -GroupScope DomainLocal -Path
'OU=Groups,OU=District,DC=ad,DC=example,DC=com'
Set-LapsADReadPasswordPermission -Identity
'OU=Workstations,OU=District,DC=ad,DC=example,DC=com' `
  -AllowedPrincipals 'AD\LAPS Readers - Workstations'
```

Check who can read: `Find-LapsADExtendedRights -Identity 'OU=Workstations,OU=District,DC=ad,DC=example,DC=com'`.

14.3 Configure the policy

GPO `Computer - Windows LAPS` linked to the workstation and server OUs. Path: Computer Configuration → Policies → Administrative Templates → System → **LAPS**.

SETTING	RECOMMENDED VALUE
Configure password backup directory	Enabled — *Active Directory*
Password Settings	Complexity: large + small letters + numbers + specials (or a passphrase option); length 20; age 30 days
Enable password encryption	Enabled
Configure authorized password decryptors	`AD\LAPS Readers - Workstations` (default is Domain Admins)
Name of administrator account to manage	Leave not configured to manage the built-in Administrator; set a name if you use a custom local admin
Configure automatic account management (2025 and Windows 11 24H2+)	Optional: let LAPS create/rename/disable the managed account
Post-authentication actions	Reset the password and log off the managed account; grace period 8 hours
Enable password backup for DSRM accounts	In a separate GPO linked to *Domain Controllers* (Section 14.5)

Windows Server 2025 member servers can also apply Microsoft's LAPS settings through OSConfig (scenario `LAPS/WindowsServer/2025/MemberServer`, member servers only).

14.4 Retrieve, rotate and verify

GUI: ADUC → open the computer object → *LAPS* tab → **Show password** / **Expire now**. Windows Admin Center also shows it.

```
Invoke-LapsPolicyProcessing                          # on the client: apply policy now
Get-LapsADPassword -Identity CLIENT01 -AsPlainText  # read (authorized users only)
Get-LapsADPassword -Identity CLIENT01 -IncludeHistory
Reset-LapsPassword                                   # on the client: rotate immediately
Set-LapsADPasswordExpirationTime -Identity CLIENT01 # make it rotate at next policy cycle
```

Verify on a client: Event Viewer → Applications and Services Logs → Microsoft → Windows → *LAPS* → Operational shows a successful password update; `Get-LapsADPassword` returns a recent *PasswordUpdateTime*.

14.5 DSRM passwords on domain controllers

Windows LAPS can also manage each DC's DSRM password (requires encryption and domain functional level 2016). Create GPO `DC - Windows LAPS DSRM` linked to *Domain Controllers* with *Configure password backup directory* = Active Directory, *Enable password encryption*, and *Enable password backup for DSRM accounts*. Retrieve with `Get-LapsADPassword -Identity DC01 -AsPlainText`. Keep an offline copy procedure for forest recovery, when AD itself is unavailable.

14.6 Troubleshooting

- **No password in AD** → computer's OU lacks self-permission, or GPO not applied (`gpresult /h`). LAPS events explain failures.
- **"Access denied" reading** → the reader is not in the authorized decryptors group, or lacks read permission on the OU.
- **Legacy LAPS still installed** → both may fight over the same account; remove the legacy CSE once migrated.

Security: audit password reads (event 4662 on DCs with Directory Service Access auditing), keep reader groups small, and separate readers for workstations, servers and DCs.

Rollback: unlink the GPO. Passwords stay as last set — retrieve them first, or set a known password per machine.

Practical lab 14 — Deploy Windows LAPS (Intermediate, 45 min)

VMs: DC01, CLIENT01 (domain-joined, in the Workstations OU). **Steps:** run 14.2; create and link the GPO in 14.3; on CLIENT01 run `gpupdate /force` and `Invoke-LapsPolicyProcessing`; read the password from DC01; sign in to CLIENT01 as `.\Administrator` with it; wait for the post-authentication action and confirm the password changed. **Cleanup:** none. **Knowledge check:** Where is the encrypted password stored? *On the computer's own AD object.* Who generates it? *The managed computer.*

15 · Security hardening

Overview. Hardening means removing what attackers use: weak defaults, unnecessary access and missing visibility. Start from Microsoft's security baseline, then add the controls below. Test every change on a pilot server first — some settings break old applications.

15.1 Apply the Windows Server 2025 security baseline

Option A — OSConfig (Windows Server 2025 only). Applies Microsoft's baseline locally and keeps it from drifting.

```
Install-Module -Name Microsoft.OSConfig -Scope AllUsers -Repository PSGallery
# Choose ONE role scenario:
Set-OSConfigDesiredConfiguration -Scenario SecurityBaseline/WindowsServer/2025/MemberServer -
Default
# Set-OSConfigDesiredConfiguration -Scenario
SecurityBaseline/WindowsServer/2025/DomainController -Default
# Set-OSConfigDesiredConfiguration -Scenario
SecurityBaseline/WindowsServer/2025/WorkgroupMember -Default
Set-OSConfigDesiredConfiguration -Scenario Defender/Antivirus/WindowsServer/2025 -Default
# Check compliance
Get-OSConfigDesiredConfiguration -Scenario SecurityBaseline/WindowsServer/2025/MemberServer |
  Format-Table Name, @{n='Status';e={$_.Compliance.Status}}, @{n='Reason';e=
{$_.Compliance.Reason}} -Wrap
```

Requires verification: early module versions used scenario names like `SecurityBaseline/WS2025/MemberServer`. Use the name format your installed module documents. A restart is required after applying.

Option B — Group Policy. Download the *Microsoft Security Compliance Toolkit*, import the Windows Server 2025 baseline GPOs into GPMC (*Import Settings*), and link them to pilot OUs first. Use *Policy Analyzer* from the toolkit to compare against your existing GPOs. CIS Benchmarks and DISA STIGs are stricter alternatives for regulated environments.

> **PRODUCTION IMPACT** Do not mix OSConfig and conflicting GPO settings on the same server — Group Policy wins when it is applied, and the two will fight over values. Choose one method per server group.

15.2 Microsoft Defender Antivirus

```
Get-MpComputerStatus | Select-Object AMServiceEnabled, RealTimeProtectionEnabled,
AntivirusSignatureLastUpdated
Update-MpSignature
Get-MpPreference | Select-Object ExclusionPath    # keep exclusions minimal and documented
```

Windows Server roles get automatic exclusions; do not add broad folder exclusions without vendor guidance. Attack Surface Reduction rules and tamper protection are managed through Microsoft Defender for Endpoint or Intune.

15.3 SMB security

```
Get-SmbServerConfiguration | Select-Object EnableSMB1Protocol, RequireSecuritySignature,
EncryptData, RejectUnencryptedAccess
Get-SmbClientConfiguration | Select-Object RequireSecuritySignature, EnableInsecureGuestLogons
Get-WindowsOptionalFeature -Online -FeatureName SMB1Protocol      # should be Disabled
# Require signing on file servers (inbound), as already default on DCs
Set-SmbServerConfiguration -RequireSecuritySignature $true -Force
# Encrypt a sensitive share
Set-SmbShare -Name 'HR$' -EncryptData $true -Force
```

On Windows Server 2025, outbound signing is required by default and guest access is blocked. If an older NAS or copier stops working, update its firmware or enable SMB signing on it — avoid turning signing off on Windows. If you must, scope the exception and document it.

15.4 Credential protection

- **Virtualization-based security (VBS) and Credential Guard** isolate credentials from the OS. On Windows Server 2025, Credential Guard is enabled by default on eligible domain-joined member servers (not DCs) whose hardware supports VBS. Check:

  ```
  (Get-CimInstance -Namespace root\Microsoft\Windows\DeviceGuard -ClassName
  Win32_DeviceGuard).SecurityServicesRunning
  # 1 = Credential Guard running, 2 = HVCI (memory integrity) running
  ```

- **LSA protection** runs LSASS as a protected process so tools cannot read its memory. Enable via Group Policy: Computer Configuration → Administrative Templates → System → Local Security Authority → *Configure LSASS to run as a protected process* = Enabled with UEFI Lock. Test with third-party authentication or smart-card software first. *Requires verification:* default state varies by install type.
- **Protected Users** group: add admin accounts (not service accounts). Members cannot use NTLM, DES/RC4 or unconstrained delegation, and their credentials are not cached.

- **NTLM:** NTLMv1 is gone; plan NTLM reduction by enabling *Network security: Restrict NTLM: Audit...* policies, reviewing event logs, then blocking where possible.

15.5 BitLocker

```
Install-WindowsFeature BitLocker -IncludeAllSubFeature -IncludeManagementTools -Restart
Enable-BitLocker -MountPoint 'C:' -TpmProtector -EncryptionMethod XtsAes256 -UsedSpaceOnly
Add-BitLockerKeyProtector -MountPoint 'C:' -RecoveryPasswordProtector
$kp = (Get-BitLockerVolume -MountPoint 'C:').KeyProtector | Where-Object KeyProtectorType -eq
'RecoveryPassword'
Backup-BitLockerKeyProtector -MountPoint 'C:' -KeyProtectorId $kp.KeyProtectorId
```

Enable *Store BitLocker recovery information in Active Directory Domain Services* by policy first so the last command succeeds. BitLocker on physical DCs and branch servers protects AD data if the hardware is stolen. Store recovery keys for DCs outside AD as well.

15.6 Remote access

- RDP with NLA only; scope firewall rules to admin subnets; no direct internet exposure — use VPN or RD Gateway with MFA.
- Prefer PowerShell remoting and Windows Admin Center over interactive RDP sign-ins; consider WinRM over HTTPS (5986) across untrusted segments.
- Deny high-privilege accounts (Domain Admins) the right to sign in to workstations and member servers: User Rights Assignment → *Deny log on locally* / *Deny log on through Remote Desktop Services* in workstation GPOs.

15.7 PowerShell logging

Computer Configuration → Administrative Templates → Windows Components → Windows PowerShell → *Turn on PowerShell Script Block Logging* (Enabled) and *Turn on Module Logging* (modules: `*`). Events appear in Microsoft-Windows-PowerShell/Operational (event 4104). Use *AllSigned* or *RemoteSigned* execution policy with signed admin scripts.

15.8 Active Directory-specific protections

- Keep Domain Admins, Enterprise Admins and Schema Admins nearly empty (Schema Admins empty except during schema changes). Review monthly: `Get-ADGroupMember 'Domain Admins' -Recursive`.

- Restrict who can create **delegated Managed Service Accounts (dMSA)** or create objects in OUs; 2025 dMSA permissions have been abused in published attacks. Apply current security updates and audit OU *Create child* permissions.
- Find accounts with unconstrained delegation: `Get-ADComputer -Filter {TrustedForDelegation -eq $true}` — only DCs should appear.
- Find stale accounts: `Search-ADAccount -AccountInactive -TimeSpan 90.00:00:00 -UsersOnly`.
- Require LDAP signing and LDAP channel binding on DCs after auditing (events 2887/2889 on DCs show clients using unsigned binds).

15.9 Hardening checklist

✓	CONTROL	VERIFY WITH
☐	Security baseline applied (OSConfig or GPO) and compliant	`Get-OSConfigDesiredConfiguration` / Policy Analyzer
☐	Firewall on, all profiles; management rules scoped; logging on	`Get-NetFirewallProfile`
☐	Defender real-time protection on, signatures current	`Get-MpComputerStatus`
☐	SMB1 disabled; signing required; guest access off	`Get-SmbServerConfiguration`
☐	Secure Boot on; TPM 2.0 present; BitLocker on physical servers	`Confirm-SecureBootUEFI`, `Get-BitLockerVolume`
☐	Credential Guard / LSA protection where supported	Win32_DeviceGuard
☐	Windows LAPS managing local Administrator	`Get-LapsADPassword`
☐	Separate admin accounts; privileged groups reviewed; admins in Protected Users	`Get-ADGroupMember`
☐	RDP: NLA, no internet exposure	Firewall/router review
☐	Advanced audit policy and PowerShell logging on; logs forwarded	`auditpol /get /category:*`
☐	Updates current; monthly patch process followed	`Get-HotFix | Sort InstalledOn -Desc`
☐	Backups include system state; offline/immutable copy; restore tested	Chapter 17

16 · Active Directory maintenance & troubleshooting

Overview. Every DC holds a full writable copy of the directory and replicates changes to the others (multi-master). Check replication and DC health weekly, and after any change to DCs, DNS or sites.

16.1 Check and force replication (GUI)

1. **Tools** → *Active Directory Sites and Services* → *Sites* → *Default-First-Site-Name* → *Servers* → expand the destination DC → *NTDS Settings*.
2. Right-click a connection object in the right pane → *Replicate Now*. This replicates only from that partner to this DC — a safe, targeted action. A confirmation appears; an error message names the problem.

16.2 Repadmin and Dcdiag

COMMAND	WHAT IT DOES
`repadmin /replsummary`	One-screen health summary: largest delta and failures per DC, as source and destination. Start here.
`repadmin /showrepl DC01`	Last inbound replication attempt for each partition and partner, with error text.
`repadmin /queue`	Inbound replication requests still waiting.
`repadmin /syncall DC01 /AdeP`	Synchronize DC01 with all partners, all partitions, across sites, push mode. Useful after fixing a fault; avoid running routinely in large environments.
`repadmin /kcc`	Recalculate replication topology now (e.g. after adding a DC or site link).
`repadmin /replicate DC02 DC01 "DC=ad,DC=example,DC=com"`	Replicate one partition from DC01 to DC02 immediately.
`dcdiag /v /c /e /q`	All tests, all DCs in the forest, errors only. Run on a DC.
`dcdiag /test:dns /v`	DNS registration and resolution tests.

PowerShell equivalents: `Get-ADReplicationPartnerMetadata -Target * -Scope Domain`, `Get-ADReplicationFailure -Target DC01`, `Sync-ADObject`.

Common replication errors: *1722 RPC server unavailable* → network/firewall or DNS; *8524 DNS lookup failure* → partner's records missing; *1311* → site link topology; *8606/8614* → a DC was offline longer than the tombstone lifetime (180 days) — it must be removed and rebuilt, not forced back.

16.3 Remove a failed (dead) domain controller — metadata cleanup

> **DESTRUCTIVE** Only remove a DC that is permanently offline and will never return. If it holds FSMO roles, seize them first (16.4). If the DC is merely offline for repair, do not clean it up.

Correction: the 2022 edition used ADSI Edit and deleted objects under *File Replication Service*. FRS is not used by supported domains, and since Windows Server 2008 deleting the DC through the normal consoles performs metadata cleanup automatically.

1. **ADUC** → *Domain Controllers* OU → right-click the dead DC → *Delete* → *Yes* → tick ***This Domain Controller is permanently offline and can no longer be demoted...*** → **Delete**. If it was a Global Catalog, confirm the next prompt.
2. **Sites and Services** → *Sites* → its site → *Servers* → right-click the server object → *Delete* (its NTDS Settings should already be gone; if not, delete NTDS Settings first and tick *Use Delete Subtree server control*).
3. **DNS** → remove its A record, its name-server (NS) records in each zone, its GUID CNAME under `_msdcs`, and any SRV records in `_msdcs`, `_sites`, `_tcp`, `_udp`.
4. Remove its IP from DHCP option 006 and from other servers' DNS client settings.
5. Verify: `repadmin /replsummary` no longer lists it; `dcdiag /e /q` is clean.

Command-line fallback: `ntdsutil "metadata cleanup" "remove selected server CN=DC03,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration,DC=ad,DC=example,DC=com" q q`.

16.4 FSMO roles — transfer vs seize

Five operations master (FSMO) roles exist: Schema Master and Domain Naming Master (one per forest); PDC Emulator, RID Master and Infrastructure Master (one per domain).

	TRANSFER	SEIZE
When	Both DCs online (planned maintenance, decommissioning)	Current holder permanently failed
Safe?	Yes — roles hand over cleanly	Last resort. The old holder must **never** come back online; wipe it.
Rights	Domain Admins for domain roles; Enterprise Admins for forest roles; Schema Admins too for Schema Master	

```
# Who holds what
netdom query fsmo
Get-ADDomain | Select-Object PDCEmulator, RIDMaster, InfrastructureMaster
Get-ADForest | Select-Object SchemaMaster, DomainNamingMaster

# Transfer all five roles to DC02 (both DCs online)
Move-ADDirectoryServerOperationMasterRole -Identity DC02 `
  -OperationMasterRole PDCEmulator, RIDMaster, InfrastructureMaster, SchemaMaster,
DomainNamingMaster

# Seize (holder is permanently gone) — same command with -Force
Move-ADDirectoryServerOperationMasterRole -Identity DC02 -OperationMasterRole PDCEmulator,
RIDMaster -Force
```

Ntdsutil method (from the 2022 edition, still valid): `ntdsutil` → `roles` → `connections` → `connect to server DC02` → `q` → `seize pdc` / `seize rid master` / `seize infrastructure master` / `seize naming master` / `seize schema master` (use `transfer …` instead of `seize` when the holder is online) → `q` → `q`. Ntdsutil tries a transfer before seizing.

After moving the PDC Emulator, reconfigure time (Chapter 8). **Verify:** `netdom query fsmo`, then `dcdiag /test:knowsofroleholders /test:fsmocheck`.

16.5 Reset the DSRM Administrator password

```
ntdsutil "set dsrm password" "reset password on server null" q q     # prompts for the new
password (local DC)
ntdsutil "set dsrm password" "sync from domain account dsrm-sync" q q # alternative: copy from
a disabled domain account
```

Better: let Windows LAPS manage DSRM passwords (Section 14.5).

17 · Backup & recovery

Overview. A backup only counts if you have restored from it. Follow the 3-2-1 rule: three copies, on two kinds of media, one off-site — and at least one copy **offline or immutable** so ransomware cannot delete it.

17.1 Windows Server Backup

```
Install-WindowsFeature Windows-Server-Backup -IncludeManagementTools
```

Manual backup (GUI): Tools → *Windows Server Backup* → *Local Backup* → *Backup Once…* → *Different options* → *Custom* → **Add Items**: *System state*, *Bare metal recovery* and data volumes → destination (dedicated disk or remote share) → **Backup**.

Scheduled backup: *Backup Schedule…* → *Custom* → items → time(s) → *Back up to a hard disk that is dedicated for backups* (recommended; the disk is reformatted and hidden) or a network share (keeps only the latest backup).

```
wbadmin start systemstatebackup -backupTarget:E: -quiet          # system state only (DCs:
includes AD and SYSVOL)
wbadmin start backup -backupTarget:E: -include:C:,D: -allCritical -systemState -quiet
wbadmin get versions                                              # list backups
```

Back up every DC's system state at least daily. A backup older than the tombstone lifetime (180 days) cannot be used to restore AD.

17.2 Restore files and volumes

Windows Server Backup → *Recover…* → *This server* → choose the date → *Files and folders* or *Volumes* → items → *Another location* (safest) or original location → **Recover**. For a full rebuild, boot from installation media → *Repair* → *Troubleshoot* → *System Image Recovery*.

17.3 Active Directory Recycle Bin

Lets you undelete users, groups and OUs with all attributes and memberships, without a backup. Enabling it is irreversible and requires Enterprise Admins.

```
Enable-ADOptionalFeature -Identity 'Recycle Bin Feature' -Scope ForestOrConfigurationSet -
Target 'ad.example.com'
# Restore a deleted user
Get-ADObject -Filter 'SamAccountName -eq "jane.smith"' -IncludeDeletedObjects | Restore-
ADObject
```

GUI: ADAC → domain → *Deleted Objects* → right-click → *Restore*. Restore parent OUs before their children.

17.4 Restoring a domain controller

- **Non-authoritative restore** (one DC is damaged, others healthy): usually simplest to demote/clean up and promote a new DC. Alternatively restore system state in DSRM; the DC then receives newer changes from partners.
- **Authoritative restore** (objects deleted and Recycle Bin not enabled): restore system state in DSRM, then before restarting run `ntdsutil "activate instance ntds" "authoritative restore" "restore subtree OU=Faculty,OU=Users,OU=District,DC=ad,DC=example,DC=com" q q`, restart, and let it replicate out.
- **Forest recovery** (all DCs lost, e.g. ransomware): follow Microsoft's *Active Directory Forest Recovery Guide* in an isolated network. Practise it in a lab before you need it.

> **DESTRUCTIVE** System state restores and authoritative restores overwrite directory data. Follow Microsoft's procedure exactly, in DSRM, and document every step.

17.5 DHCP and DNS recovery

DHCP: Section 7.6 (`Export-DhcpServer` / `Import-DhcpServer`, or failover partner takes over). DNS: AD-integrated zones are restored with AD; for file-based zones copy `%SystemRoot%\System32\dns`. Export each zone for documentation: `Export-DnsServerZone -Name 'ad.example.com' -FileName 'ad.example.com.bak'`.

17.6 Recovery test checklist (quarterly)

1. Restore a file from last night's backup to an alternate location.
2. Restore a deleted test user from the AD Recycle Bin.
3. Restore a DC system state into an isolated lab VM and run `dcdiag`.
4. Import the DHCP export on a lab server.
5. Confirm the offline/immutable copy exists and is readable.
6. Record times achieved against your Recovery Time and Recovery Point Objectives.

Appendix A · PowerShell & command quick reference

TASK	COMMAND
List / install / remove roles	`Get-WindowsFeature` · `Install-WindowsFeature <Name> -IncludeManagementTools` · `Uninstall-WindowsFeature`
Rename, join domain, restart	`Rename-Computer` · `Add-Computer -DomainName` · `Restart-Computer`
IP configuration	`Get-NetIPConfiguration` · `New-NetIPAddress` · `Set-DnsClientServerAddress`
Connectivity	`Test-NetConnection <host> -Port <n>` · `Resolve-DnsName`
Firewall	`Get-NetFirewallProfile` · `Get-NetFirewallRule` · `New-NetFirewallRule`
Services / processes	`Get-Service` · `Restart-Service` · `Get-Process`
Event logs	`Get-WinEvent -LogName System -MaxEvents 50`
Remote session	`Enter-PSSession DC01` · `Invoke-Command -ComputerName DC01 -ScriptBlock { … }`
Users and groups	`Get-ADUser` · `New-ADUser` · `Set-ADAccountPassword` · `Unlock-ADAccount` · `Add-ADGroupMember`
Find locked / inactive accounts	`Search-ADAccount -LockedOut` · `Search-ADAccount -AccountInactive -TimeSpan 90.00:00:00`
Domain / DC info	`Get-ADDomain` · `Get-ADForest` · `Get-ADDomainController -Filter *` · `nltest /dsgetdc:ad.example.com`
Replication & health	`repadmin /replsummary` · `repadmin /showrepl` · `dcdiag /e /q`
FSMO	`netdom query fsmo` · `Move-ADDirectoryServerOperationMasterRole`

TASK	COMMAND
Group Policy	`gpupdate /force` · `gpresult /h report.html` · `Get-GPO -All` · `Backup-GPO` · `Get-GPOReport -All -ReportType Html -Path gpos.html`
DNS	`Get-DnsServerZone` · `Add-DnsServerResourceRecordA` · `Get-DnsServerForwarder`
DHCP	`Get-DhcpServerv4Scope` · `Get-DhcpServerv4Lease` · `Get-DhcpServerv4ScopeStatistics`
Time	`w32tm /query /status` · `w32tm /resync`
Disk usage	`Get-Volume` · `Get-PSDrive -PSProvider FileSystem`
Updates installed	`Get-HotFix | Sort-Object InstalledOn -Descending`
Windows LAPS	`Get-LapsADPassword -Identity <PC> -AsPlainText`
Backup	`wbadmin get versions` · `wbadmin start systemstatebackup -backupTarget:E:`

Appendix B · Network port reference

SERVICE	PORT(S)	NOTES
DNS	53 TCP/UDP	Clients → DCs; DCs → forwarders
Kerberos / password change	88 TCP/UDP · 464 TCP/UDP	Clients → DCs
NTP (Windows Time)	123 UDP	Clients → DCs; PDC emulator → external
RPC endpoint mapper + dynamic	135 TCP · 49152–65535 TCP	AD replication, management, DFSR
LDAP / LDAPS	389 TCP/UDP · 636 TCP	LDAPS needs a DC certificate
Global Catalog / GC over TLS	3268 · 3269 TCP	
SMB (file shares, SYSVOL, NETLOGON)	445 TCP	Never expose to the internet
SMB over QUIC	443 UDP	Optional; needs certificate
AD Web Services (PowerShell AD module, ADAC)	9389 TCP	
DHCP / DHCP failover	67–68 UDP · 647 TCP	Router needs a relay (IP helper) per VLAN
Remote Desktop	3389 TCP/UDP	Scope to admin networks
WinRM / PowerShell remoting	5985 TCP (HTTP) · 5986 TCP (HTTPS)	Server Manager, WAC
WSUS	8530 TCP (HTTP) · 8531 TCP (HTTPS)	Clients → WSUS
Windows Admin Center gateway	443 TCP (or the port chosen at install)	*Requires verification* for your WAC version
NetBIOS / WINS (legacy)	137–139 · 42	Avoid; see Appendix D

Appendix C · Installation checklist

✓	BEFORE INSTALLATION
☐	Hardware on vendor's Windows Server 2025 certified list; firmware/BIOS updated
☐	UEFI mode, Secure Boot on, TPM 2.0 enabled; RAID configured; storage driver available
☐	Edition, licensing (cores, CALs) and activation method decided
☐	Name, IP, subnet, gateway, DNS servers and OU documented
☐	Bootable USB created and tested (Chapter 3)

✓	AFTER INSTALLATION
☐	Strong Administrator password stored in vault
☐	Renamed; static IP; time zone; activated
☐	Fully updated; firewall on; Remote Desktop with NLA (if needed)
☐	Domain-joined and moved to the correct OU; LAPS password present
☐	Baseline applied; Defender healthy; included in backup and monitoring
☐	Server documented (asset list, roles, owners)

The security hardening checklist is in Section 15.9; the recovery test checklist in Section 17.6.

Appendix D · Legacy technology reference

These procedures from the 2022 edition are kept for reference only. Do not use them for new deployments.

WINS

WINS resolved NetBIOS names before DNS was universal. It is deprecated; Windows Server 2025 is the last release that includes it (supported through November 2034). Before removing it: check WINS for registrations from devices that do not use DNS, look for applications configured with single-label names, then use DNS suffix search lists (DHCP option 119 or GPO) or a *GlobalNames* zone for single-label names. Remove DHCP options 044/046 and WINS addresses from adapters.

Original procedure (summary): Add Roles and Features → *WINS Server* feature → in WINS Manager add replication partners for each WINS server → add WINS addresses to every server adapter and DHCP option 044, node type 0x8 in option 046.

Disabling IPv6 (DisabledComponents = 0xFFFFFFFF)

Replaced by Section 4.4. If found on existing servers, change the value to `0x20` (prefer IPv4) or delete it, then restart and test.

File Replication Service (FRS) for SYSVOL

Removed since Windows Server 2019; 2025 DCs cannot join a domain still using FRS. Check with `dfsrmig /getglobalstate`; if not *Eliminated*, migrate using `dfsrmig /setglobalstate 1` → `2` → `3`, waiting for every DC to reach each state (`dfsrmig /getmigrationstate`) — from a supported older DC, before adding 2025 DCs.

VBScript logon scripts and drive renaming

Replaced by Group Policy Preferences (Section 10.6). The original `rename-mapped-drives.vbs` and the `IF "%OS%"=="Windows_NT"` branch for Windows 9x clients are obsolete.

dsadd with shared default passwords

Replaced by the PowerShell script in Section 10.5. `dsadd` still exists but shared default passwords are insecure.

SMB1, NTLMv1, TLS 1.0/1.1

SMB1 is not installed by default and should stay that way; NTLMv1 is removed; TLS 1.0/1.1 are disabled by default. Replace devices that need them.

Appendix E · Troubleshooting decision trees

Users cannot sign in (authentication failure)

1. One user or many? **One** → check lockout and expiry: `Get-ADUser jane.smith -Properties LockedOut,PasswordExpired,Enabled` → unlock or reset.
2. **Many on one computer** → *"trust relationship failed"*: `Test-ComputerSecureChannel -Repair -Credential AD\admin.jsmith`.
3. **Many everywhere** → can clients find a DC? `nltest /dsgetdc:ad.example.com` → if not, DNS (Chapter 6).
4. Clock skew over 5 minutes? `w32tm /stripchart /computer:DC01 /samples:3` → fix time (Chapter 8).
5. Kerberos errors in the client System log (KDC events) → check for duplicate SPNs: `setspn -X`; purge tickets: `klist purge`.
6. Recently added a 2025 DC? Confirm functional level 2016+, healthy replication (`repadmin /replsummary`) and current updates on all DCs.

AD replication failing

1. `repadmin /replsummary` → identify failing DC pairs and error numbers.
2. DNS: can each DC resolve the other's GUID CNAME? `dcdiag /test:dns`.
3. Network: `Test-NetConnection DC02 -Port 135` and 389, 445; check firewalls between sites.
4. Time within 5 minutes?
5. Error 8606/8614 (tombstone) → remove and rebuild the stale DC; do not force replication.
6. Fixed → `repadmin /syncall /AdeP` and recheck.

Domain controller promotion fails

1. DNS on the new server points to an existing DC? `nltest /dsgetdc:` works?
2. Functional level ≥ 2016 and SYSVOL on DFSR?
3. Credentials in Domain Admins (and Enterprise/Schema Admins for the first 2025 DC)?
4. Existing DCs healthy (`dcdiag`)?
5. Read `%SystemRoot%\debug\dcpromo.log` for the exact error.

DNS resolution: Section 6.6. DHCP: Section 7.6. Group Policy: Section 13.8. Windows Update: Section 9.6. Backup failures: check *Microsoft-Windows-Backup* events and target disk space; VSS writer status with `vssadmin list writers`.

Appendix F · Glossary

TERM	DEFINITION
Active Directory Domain Services (AD DS)	Microsoft's directory service: stores users, groups and computers and authenticates them in a Windows domain.
AD CS	Active Directory Certificate Services — Microsoft's certificate authority role for issuing internal certificates.
ADMX / Central Store	Administrative Template files defining Group Policy settings; the Central Store is their shared copy in SYSVOL.
Azure Arc	Service that connects on-premises servers to Azure for management, updates, monitoring and pay-as-you-go licensing.
BitLocker	Full-volume encryption built into Windows, normally protected by the TPM.
CAL	Client Access License — required for each user or device that accesses Windows Server.
Credential Guard / VBS	Uses hardware virtualization to isolate credentials from the rest of the operating system.
DFS Replication (DFSR)	Replication engine that copies SYSVOL (and other folders) between servers; replaced FRS.
DHCP	Protocol that assigns IP addresses and network settings to devices automatically.
DNS	Hierarchical naming system that translates names to IP addresses; AD uses it to locate DCs.
Domain controller (DC)	Server running AD DS that authenticates users and computers and holds a copy of the directory.
dMSA / gMSA	Delegated / group Managed Service Accounts — service accounts whose passwords Windows manages automatically.
DSRM	Directory Services Restore Mode — offline mode for repairing or restoring AD on a DC, with its own local password.

TERM	DEFINITION
Firewall	Control that allows or blocks network traffic by rules; Windows Defender Firewall runs on every server.
FSMO roles	Five operations master roles held by specific DCs (Schema, Domain Naming, PDC Emulator, RID, Infrastructure).
Functional level	Setting that determines which AD features are available and which DC versions are allowed.
Global Catalog	DC holding a partial copy of every object in the forest, used for searches and sign-in.
Group Policy / GPO	Centralized settings for computers and users, stored as Group Policy Objects linked to sites, domains or OUs.
Hotpatching	Applying security updates in memory without a restart; Windows Server 2025 via Azure Arc subscription.
IP address	Numeric label identifying a device on a network (IPv4 e.g. 192.0.2.10; IPv6 e.g. 2001:db8::10).
JEA	Just Enough Administration — PowerShell endpoints that allow only specific admin tasks.
Kerberos	Default ticket-based authentication protocol in AD.
LAPS	Local Administrator Password Solution — unique, rotating local admin passwords stored in AD or Entra ID.
LDAP / LDAPS	Protocol for querying directories; LDAPS is LDAP over TLS.
NTLM	Older challenge-response authentication protocol; deprecated in favour of Kerberos.
Organizational unit (OU)	Container in AD for organizing objects, linking GPOs and delegating administration.
OSConfig	PowerShell-based security configuration tool for applying the Windows Server 2025 baseline with drift control.
OSE	Operating System Environment — a physical or virtual OS instance, for licensing.

TERM	DEFINITION
Protected Users	Built-in AD group whose members get stronger credential protections.
RAID	Combining multiple disks for redundancy and/or performance.
Security group	AD group used to grant permissions; distribution groups are for email only.
Server Core	Installation option without the desktop; managed by command line or remotely.
Server Manager	Console for managing roles and features on local and remote servers.
SConfig	Menu-driven configuration tool for basic server settings.
SMB signing / encryption	Protections for file-sharing traffic against tampering (signing) and eavesdropping (encryption).
SYSVOL	Shared folder on every DC holding Group Policy files and logon scripts (NETLOGON), replicated by DFSR.
TPM	Trusted Platform Module — hardware chip that stores keys and measures boot integrity.
UEFI / Secure Boot	Modern firmware interface; Secure Boot allows only trusted boot loaders to run.
Virtualization	Running multiple virtual machines on one physical host (Hyper-V on Windows Server).
Windows Admin Center (WAC)	Browser-based management tool for Windows Server.
WSUS	Windows Server Update Services — on-premises update distribution; deprecated but supported.

Appendix G · References

Primary sources (Microsoft Learn unless noted). Review date for this edition: October 2026. Re-check before each revision; URLs change.

- What's new in Windows Server 2025 — learn.microsoft.com/windows-server/get-started/whats-new-windows-server-2025
- Features removed or no longer developed in Windows Server — learn.microsoft.com/windows-server/get-started/removed-deprecated-features-windows-server
- Hardware requirements for Windows Server — learn.microsoft.com/windows-server/get-started/hardware-requirements
- Comparison of Windows Server editions — learn.microsoft.com/windows-server/get-started/editions-comparison
- Windows Server 2025 lifecycle — learn.microsoft.com/lifecycle/products/windows-server-2025
- Forest and domain functional levels — learn.microsoft.com/windows-server/identity/ad-ds/active-directory-functional-levels
- Lower domain and forest functional levels — learn.microsoft.com/windows-server/identity/ad-ds/plan/lower-domain-forest-functional-levels
- Selecting the forest root domain — learn.microsoft.com/windows-server/identity/ad-ds/plan/selecting-the-forest-root-domain
- Guidance for configuring IPv6 in Windows — learn.microsoft.com/troubleshoot/windows-server/networking/configure-ipv6-in-windows
- SMB security hardening; Control SMB signing behavior — learn.microsoft.com/windows-server/storage/file-server/smb-security-hardening
- Windows LAPS overview — learn.microsoft.com/windows-server/identity/laps/laps-overview
- Deploy Windows Server 2025 security baselines with OSConfig — learn.microsoft.com/windows-server/security/osconfig/osconfig-how-to-configure-security-baselines
- Windows Server Update Services documentation — learn.microsoft.com/windows-server/administration/windows-server-update-services/get-started/windows-server-update-services-wsus
- Windows Time service — learn.microsoft.com/windows-server/networking/windows-time-service/windows-time-service-top
- Transfer or seize FSMO roles — learn.microsoft.com/troubleshoot/windows-server/active-directory/transfer-or-seize-operation-master-roles-in-ad-ds

- Active Directory Forest Recovery Guide — learn.microsoft.com/windows-server/identity/ad-ds/manage/ad-forest-recovery-guide
- KB5071836 — WINS removal: moving forward with modern name resolution (support.microsoft.com)
- NIST SP 800-63B Rev. 4, Digital Identity Guidelines: Authentication — pages.nist.gov/800-63-4/sp800-63b.html
- CIS Microsoft Windows Server Benchmarks — cisecurity.org/benchmark/microsoft_windows_server

Appendix H · Traceability: 2022 edition → this guide

2022 SECTION	STATUS	WHERE NOW / WHAT CHANGED
Requirements	Modified	2.1 — new CPU instructions, TPM/UEFI, realistic sizes
Glossary	Expanded	Appendix F — SYSVOL updated (DFSR), modern terms added
Virtualization rights; Licensing editions	Valid w/ modifications	2.2–2.3 — Azure Edition, pay-as-you-go, M365 admin center
Pre-installation & installation	Modified	3.4 — new Setup screens; size formula corrected
Bootable USB (GUI / PS)	Rewritten	3.2–3.3 — Secure Boot kept on; WIM split
Server initial configuration	Modified	4.1–4.12
Product activation (SLMGR)	Valid w/ modifications	4.6 — /ato, /dlv; KMS/ADBA/AVMA; VBScript note
Disable IPv6 via registry	Insecure / not recommended	4.4 prefer IPv4 (0x20); old method → Appendix D
Disable Windows Firewall	Insecure	4.8 — keep on, scope rules, logging
Turn off IE ESC	Not recommended	4.11 — no browsing on servers
Domain services & AD setup	Modified	5 — namespace (no .local), 2025 functional levels, PowerShell, second DC
Reverse lookup zones; scavenging; forwarders; DIS/OpenDNS	Valid w/ modifications	6.1–6.4 — secure updates, one scavenging server, DIS values marked organization-specific
DHCP installation & configuration	Valid w/ modifications	7.1–7.4 — WINS options removed; DNS credentials; conflict detection
WINS installation	Deprecated	Appendix D

2022 SECTION	STATUS	WHERE NOW / WHAT CHANGED
WSUS; configuring WSUS; WSUS GPO	Deprecated (supported)	9 — requirements fixed, staged rings replace auto-approve, HTTPS
Basic AD structure for K12	Valid w/ modifications	10.1–10.3 — graduation-year OUs, redircmp, AGDLP
Shares & home directories	Modified	11.1 — Microsoft home-folder ACLs, ABE
User template; new user from template	Valid	10.4
Batch file mass import (dsadd/Excel)	Insecure	10.5 — PowerShell with unique passwords
Logon scripts – batch / VBScript	Legacy	10.6 GPP Drive Maps; Appendix D
Shadow copies	Valid w/ modifications	11.2 — XP references removed; not-a-backup warning
Volume quotas; FSRM quotas	Valid w/ modifications	11.3 — no restart needed; PowerShell; file screens
Fine-grained password policies (ACT-723)	Corrected	12 — history (since 2008), unique precedence, NIST context
Retain security log 90 days; auto-backup logs	Modified	13.2 — size + forwarding instead of retain-by-days
Security event auditing	Modified	13.1 — Advanced Audit Policy
Logon banner; screen saver lock	Valid w/ modifications	13.3–13.4 — separate GPOs; machine inactivity limit
Folder redirection	Valid w/ modifications	13.5 — legacy Windows 2000 option dropped
Restrict computers to faculty	Valid w/ modifications	13.6 — Authenticated Users Read; klist tip
gpupdate; GPMC update; RSoP; GPRESULT	Valid (dates corrected)	13.7–13.8
Disable Shutdown Event Tracker	Valid w/ modifications	4.12 — via Group Policy; keep on recommended

2022 SECTION	STATUS	WHERE NOW / WHAT CHANGED
Set time source to DIS/NTP	Valid w/ modifications	8 — /update, 0x8 flags; DIS values flagged for verification
Check replication (GUI / repadmin); dcdiag	Valid	16.1–16.2
Delete dead DC (ADSI Edit, FRS)	Partly unsupported	16.3 — console-based metadata cleanup; FRS step removed
Manually seize FSMO roles	Valid w/ modifications	16.4 — transfer vs seize, PowerShell, warnings
Reset DSRM password	Valid	16.5 + LAPS DSRM
Windows Server Backup (install, manual, scheduled, restore)	Valid	17.1–17.2 + system state, Recycle Bin, AD restore
Backup DHCP scopes (netsh)	Legacy method	7.6 — Export/Import-DhcpServer
Microsoft lifecycle dates	Corrected	2.4

Appendix I · Modernization status & expansion plan

This edition modernizes every subject in the 2022 manual and adds Windows LAPS, hardening, Server Core basics and AD recovery. The following chapters from the full modernization specification are planned for the next editions. Each will follow the standard chapter structure (overview → prerequisites → architecture → installation → configuration → verification → troubleshooting → security → rollback → lab → knowledge check → references).

#	PLANNED CHAPTER	STATUS
18	Privileged access management (tiering, PAWs, JEA, break-glass, CyberArk integration)	Partly covered in 15.4–15.8
19	Kerberos, NTLM, LDAP signing/channel binding, delegation	Partly covered in 15.4, 15.8
20	Service accounts: sMSA, gMSA, dMSA, KDS root key	Not started
21	Active Directory Certificate Services (PKI, templates, LDAPS)	Not started
22	AD Sites and Services; multisite design	Not started
23	Advanced Group Policy (WMI filters, delegation, modeling)	Partly covered in 13
24	Hyper-V	Not started
25	Failover clustering	Not started
26	Storage: ReFS, Storage Spaces / S2D, Storage Replica, deduplication	Not started
27	SMB security in depth (QUIC, auditing, dialect control)	Partly covered in 15.3
28	PowerShell automation & reusable script library (inventory, AD health, DNS, DHCP, GPO, updates, disk, services, security events, backup)	Not started
29	Windows Admin Center	Not started
30	Secure remote administration (RD Gateway, WinRM HTTPS, JEA)	Partly covered in 4.7, 15.6
31	Event monitoring, Windows Event Forwarding, SIEM (Sentinel, Splunk)	Partly covered in 13.1–13.2
32	Performance monitoring & capacity planning	Not started

#	PLANNED CHAPTER	STATUS
33	Vulnerability management	Not started
34	Disaster recovery planning (RTO/RPO, ransomware, runbooks, full forest-recovery lab)	Partly covered in 17
35	DNS security (DNSSEC, policies, auditing)	Not started
36	Remote Desktop Services	Not started
37	IIS web server	Not started
38	Windows containers	Not started
39	Azure hybrid (Arc, Monitor, Backup, Update Manager, Policy)	Not started
—	Authentic Windows Server 2025 screenshots for all GUI procedures	Placeholders marked
—	Lab execution of every procedure and script	Not started — all items documentation-verified only

End of document.